

Mining Meaning: Measurement Error in AI-Assisted Literature Reviews*

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Abstract

Researchers increasingly use generative AI, particularly large language models (LLMs), to automate tasks across the research pipeline. We study the reliability of these tools at the reading, classification, and synthesis of large bodies of academic literature. We frame LLM-assisted literature reviews as a measurement problem, treating models as measurement systems and tracing how their errors affect downstream conclusions. As a test case, we use three different implementations of ChatGPT to identify and extract metadata from economics papers that use rainfall as an instrumental variable. We benchmark each implementation against a subset of human-labeled evaluation data, and then deploy those implementations to extract metadata from the full corpus. The LLMs perform well on binary classification, but performance deteriorates as tasks demand greater contextual interpretation. More importantly, how much researchers can rely on model outputs depends not only on the complexity of the reading task but also on the type of claims the data is asked to support. The same amount of measurement error substantially affects paper-level claims while having little effect on broader claims about the literature. Measurement error in LLM-generated data is thus most consequential at precisely the level of detail that constitutes an LLM’s principal value added over human reviewers. We conclude that standard model performance metrics are informative about the quality of generated data but do not by themselves establish the credibility of downstream inference. Researchers must also evaluate whether substantive claims are robust to the measurement system used to generate the underlying data.

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1 Introduction

The rapid development of generative AI, particularly large language models (LLMs), has expanded the range of research tasks that can be delegated to machines. Researchers increasingly use LLMs for ideation (Meinke et al., 2024; Shaer et al., 2024; Si et al., 2024); data generation (Wang et al., 2023; Ma et al., 2024; Stewart and Sinha, 2025); coding (Korinek, 2023; Peng et al., 2023), annotating and classifying text (Garg and Fetzer, 2025; Briggs et al., 2026); and writing and editing (Sung et al., 2025; Wu et al., 2025). Tasks that once required substantial amounts of human labor can now be completed quickly and at a scale that was previously infeasible.

But how much can researchers rely on LLMs to streamline research workflows through the automation of labor-intensive digital tasks? Concerns exist about hallucination, context loss, operational consistency, reproducibility, and the ability to use AI-generated data to make accurate scientific inference. Yet we know relatively little about when and where a model might succeed or fail and how to interpret those successes or failures. A model may fail to find information that is present (Type II error), report information that is not present (Type I error), or correctly recognize that a concept exists while incorrectly describing what that concept is. Further, it is not sufficient to know that a model is generally or usually accurate. Understanding how much we can rely on LLMs requires understanding the type and severity of their errors and when those errors matter for the conclusions drawn from those data.

In this paper, we examine these questions by asking LLMs to read scientific papers and generate structured data for use in a literature review. When LLMs are used to generate data, the model and its implementation together function as a measurement system. We compare the performance of these systems across research tasks that vary in the amount of contextualization and interpretation they require, from relatively simple annotation to identifying and naming economic concepts from text. We then use alternative LLM measurement systems to independently construct the same literature-review dataset and compare what we learn from each. This allows us to examine both when and where LLMs succeed or fail, as well as when errors in the data they generate matter for the conclusions researchers draw from those data.

Literature reviews provide a useful setting for this investigation as “reading” is not a single task. It can mean finding information that is explicitly stated and so relatively easy to locate, such as an article title or whether a paper contains empirical analysis. Other information requires inferring concepts from context, such as identifying the dependent variable, the source of endogeneity, or the role a particular variable plays in an econometric model. At the same time, a literature review is precisely the type of research task for which LLMs appear to have a comparative advantage. Models can potentially read tens of thousands of papers and produce a much more detailed, structured characterization of a literature than a human could.

Our test case is to understand how economists use rainfall as an instrumental variable (IV) in causal or quasi-causal studies. The rainfall-IV literature makes the reading task particularly

difficult. Information on what is endogenous, how endogeneity is addressed, what the instrument is, and how rainfall is quantified can appear in equations, tables, footnotes, or different sections of a paper, can be described using inconsistent terminology, and may only be implicitly defined. Therefore, the model must do more than find words in text. It must infer the economic role those words play. The application is also substantively important. Rainfall is used to instrument for migration, health, conflict, income, agricultural production, and many other outcomes. At the same time, the validity of the exclusion restrictions underlying these applications has come under scrutiny (Mellon, 2025).

We start by developing a simple model of measurement error that distinguishes between misclassification and semantic error. We then connect these task-level errors to the substantive estimands that researchers construct from the generated data. Empirically, we build a candidate corpus of 4,191 economics papers from which we draw a human-labeled reference sample of 439 papers. We adopt a hierarchical prompting strategy that moves from binary classification to identifying and naming the economic concepts underlying those classifications. Using OpenAI’s ChatGPT-4.1, we compare three implementation strategies: zero-shot prompting, retrieval-augmented generation (RAG), and supervised fine-tuning (SFT). We benchmark all three against a common human-labeled evaluation set and then use each measurement system to independently construct data from the full corpus. Rather than selecting a preferred implementation based on validation performance metrics, we carry all three datasets through the same substantive analysis of the rainfall-IV literature.

Our first finding is that LLMs perform well when reading requires relatively simple annotation, but performance becomes heterogeneous as we demand greater contextualization and interpretation. Across the three implementations, accuracy and sensitivity are generally high. The models also recover relatively simple metadata, such as article titles and DOIs, with high semantic similarity. This similarity in levels of overall performance conceals different types of errors. RAG tends to generate more false positives while zero-shot tends to generate more false negatives. SFT generally produces a more balanced error profile. Thus, even for tasks on which all three implementations appear highly successful, they do not necessarily get the same papers wrong. Performance of all three implementations deteriorates as reading requires a higher degree of contextual interpretation. SFT generally improves upon these more difficult extraction tasks relative to zero-shot and RAG, but all three measurement systems produce substantial semantic errors. Correctly recognizing that a concept is present does not imply that the model can correctly identify what that concept is.

Our second headline finding is that how much researchers can rely on LLM-generated data depends not just on the difficulty of the task but on what researchers ask of the generated data. Among papers classified as using a rainfall IV by at least one implementation, rainfall-IV status differs across measurement systems for more than one third of papers. Even conditional on all three systems identifying the same paper as using a rainfall IV, disagreement remains over what rainfall

measure is used, what variable is instrumented, and sometimes about whether a relevant variable exists at all. As we make broader characterizations of the literature, disagreement becomes less consequential and all three measurement systems produce essentially the same picture. Economists quantify rainfall in a great many ways and use rainfall to instrument for a wide range of socioeconomic processes, raising concerns about exclusion restriction violations as well as selection on significance.

The decline in measurement system dispersion as the estimand of interest coarsens demonstrates an important feature of LLM validation: it is estimand-dependent. Task-level validation metrics characterize the quality of the generated data. They do not, by themselves, determine the consequences of measurement error for a particular substantive estimand. At the paper-level, measurement-system choice can determine whether a paper is included in the rainfall-IV literature at all. At the relationship-level, measurement-system choice can change which rainfall measure instruments which endogenous variable. When the estimand is a broad characterization of the literature, many of these disagreements wash out. But broad characterizations of the literature is where LLMs add relatively little beyond what could be achieved by a traditional human literature review. The same amount of task-level measurement error can therefore matter a great deal for one use of the generated data and very little for another.

Our study makes three contributions. First, we contribute to the rapidly growing body of literature that uses LLMs to generate structured data for scientific research. Recent studies use machine-generated data to characterize causal claims made in economics (Garg and Fetzer, 2025) and the prevalence of publication bias in political science (Briggs et al., 2026). These studies demonstrate that carefully designed LLM workflows can produce useful data at very large scale. We ask how much researchers can rely on those data. We extend this literature by treating alternative implementations as competing measurement systems and evaluating whether substantive conclusions are robust to measurement-system choice.

Second, we connect the use of LLMs as research tools to the economics literature on measurement error, misclassification, and generated variables. An extensive literature shows that error-contaminated measures can distort descriptive statistics, coefficient estimates, and statistical uncertainty (Aigner, 1973; Bollinger, 1996; Hausman et al., 1998; Bound et al., 2001; Schennach, 2022). Recent work makes the connection to AI-generated variables explicit, showing that high predictive performance does not guarantee unbiased downstream estimates or invalid inference (Egami et al., 2023; Battaglia et al., 2024; Ludwig et al., 2025; Christensen and Hansen, 2026). Research on text as data and learned proxies similarly emphasizes that computational measures require construct-specific validation (Grimmer and Stewart, 2013; Knox et al., 2022). Our measurement error model and empirical approach allow us to distinguish measurement error in the generated data from the distortion that error induces in a substantive estimand. This distinction provides a way to move from asking whether an LLM produced good data to asking whether those data are good enough

for the inference a researcher wants to make.

Finally, we contribute to the literature that critiques the use of rainfall as an instrument (Rosenzweig and Wolpin, 2000; Deaton, 2010; Dell et al., 2014; Sarsons, 2015). Our paper most closely relates to Mellon (2025), which documents through expert human review the large number of socioeconomic variables that rainfall has been shown to predict and argues that these relationships create widespread potential violations of the exclusion restriction. Our machine-assisted review recovers the same broad concern from a much larger candidate corpus. It also reveals that the same types of endogenous variables are instrumented using many different measures of rainfall, revealing a second potential identification concern: scope for researchers to choose the rainfall measure that produces the desired first-stage results. The rainfall IV we observe may be the one selected not because of its theoretical fit or economic rationale but for its significance.

That a particular LLM succeeds at a task is sufficient to demonstrate that the model has the capacity to complete the research task. But model success is not transitive: a model that succeeds at completing task A may not succeed at task B, even when the tasks are highly similar, and a different model may not succeed at completing task A. Likewise, model failure is not transitive, nor may it be evident what model failure stems from (Argyle et al., 2025). Further, it is not obvious that what counts as success or failure is the same across context. This lack of generalizability of generative AI as a research tool makes the development of new methodologies for using that tool difficult. In the absence of a generalizable method for achieving success (“proofs”) as one would have with a new econometric estimator, it is critical to quantify and understand the modes of model success and failure so that we can know when and how to use generative AI in the research process.

2 A Simple Model of Measurement Error

Valid inference begins by defining the estimand of interest and the source of uncertainty that makes the estimator stochastic (Argyle et al., 2025). Typically the uncertainty is due to either randomness in sampling from a population or randomness in assignment to treatment (Abadie et al., 2020). Sampling uncertainty can be resolved if one is able to observe all units in the population. Design uncertainty generally cannot be eliminated because, for each unit, at least one of the potential outcomes remains unobserved (Imbens and Rubin, 2015). In each case, uncertainty arises because the observed data do not contain all the information needed to recover the estimand.

Measurement introduces an analogous source of uncertainty. If a measurement system were provably correct, meaning it could be guaranteed to return the correct target value for every admissible input, there would be no mismeasurement to contend with (Gennaro et al., 2010). Absent a proof of correctness, treating measured values as error-free amounts to assuming away the measurement problem. The relevant question is thus not only how accurately a measurement system recovers the target, but how its errors propagate into the estimand (Bound et al., 2001).

A quantitative literature review is itself an exercise in estimation. The substantive estimand

describes a defined population of papers, such as the share that uses rainfall as an IV or the distribution of variables instrumented with rainfall. Estimates depend on which papers enter the review and how their contents are measured to generate the data. Random sampling introduces variation across samples of papers, while stochastic extraction can produce different measurements of the same papers. Selective inclusion can also bias the description of the target literature.¹ The promise of using LLMs in literature reviews is that they can read far more papers than human reviewers alone and potentially the entire population of interest. Reading every paper in that population would eliminate sampling uncertainty. But it would not eliminate measurement uncertainty, which can persist even when extraction is deterministic. Data are made, not found, and model-generated labels are measurements of facts about papers, not the facts themselves. Our theoretical framework examines how errors in those measurements affect the conclusions drawn about the literature.

To fix ideas on how measurement error might manifest in data generated by LLMs, let T_{ij} be the target value for concept i in paper j and $Z_{ij}^{(m)}$ be the value generated by measurement system m . The measurement system includes the family and lineage of the model, the specific prompts, retrieval procedure, and model settings like temperature, top-p, and seed. Let $g(\cdot)$ denote the descriptive analysis of interest. Applying $g(\cdot)$ to the target values defines the substantive estimand θ , while applying it to the model-generated values produces a measurement-system-specific estimate.

For population P and observed sample S , we define $\theta_P = g(\mathbf{T}_P)$, the target estimand defined on the target data, and $\hat{\theta}_m = g(\mathbf{Z}_S^{(m)})$, an estimate of the outcome of interest which may suffer from sampling-based and measurement-based uncertainty.² We can decompose the difference between the true estimand and its estimate as:

$$\hat{\theta}_m - \theta_P = \underbrace{g(\mathbf{Z}_S^{(m)}) - g(\mathbf{T}_S)}_{\text{measurement error}} + \underbrace{g(\mathbf{T}_S) - g(\mathbf{T}_P)}_{\text{sampling error}}. \quad (1)$$

The second term generates sampling-based uncertainty which, if sampling is random, will itself be random. The first term is the measurement-based distortion of the estimand which need not be random. When $S = P$, as is possible when using LLMs to read an entire literature, the sampling error goes to zero. But the measurement error remains because in using LLMs to drive S as close to P as possible, we must rely on model generated target values instead of the target value itself, which, presumably, an expert human reader would be able to recover.

¹Selective publication creates a selection problem. Reading every published paper can characterize the published literature, but does not by itself recover the distribution of all conducted studies when publication depends on study results (Andrews and Kasy, 2019; Briggs et al., 2026).

²Throughout the paper we use the terms design-based, sampling-based, and measurement-based uncertainty. It is important to note that the uncertainty in the first two terms is generated by a stochastic mechanism. The uncertainty in the third term is a discrepancy that can arise either from a stochastic or deterministic process.

2.1 Task-Level Binary Outcomes

To characterize task-level measurement errors that give rise to measurement-induced estimand distortion, we first focus on the special case of a binary target value. The concept-level measurement error is:

$$Z_{ij} = T_{ij} \oplus E_{ij}, \quad (2)$$

where Z_{ij} remains the LLM-generated measure of the (unobserved) target value T_{ij} and E_{ij} represents the measurement error.³ In the binary case $E_{ij} = 0$ indicates no error while $E_{ij} = 1$ indicates the target value was flipped. From this accounting identity we can write the binary misclassification model for concept i :

$$\begin{pmatrix} \Pr[Z_{ij} = 1 \mid T_{ij} = 0] & \Pr[Z_{ij} = 0 \mid T_{ij} = 0] \\ \Pr[Z_{ij} = 1 \mid T_{ij} = 1] & \Pr[Z_{ij} = 0 \mid T_{ij} = 1] \end{pmatrix} = \begin{pmatrix} \alpha_i & 1 - \alpha_i \\ 1 - \beta_i & \beta_i \end{pmatrix}. \quad (3)$$

Here α_i is the false positive (FP) rate, β_i is the false negative (FN) rate, $1 - \alpha_i$ is the true negative (TN) rate (specificity), and $1 - \beta_i$ is the true positive (TP) rate (sensitivity). FPs and FNs do need not occur at the same rate.

To understand how the prevalence of FPs and FNs bias the measured values, let $p_i = \Pr(T_{ij} = 1)$ and $q_i = \Pr(Z_{ij} = 1)$. Then $q_i = (1 - \beta_i)p_i + \alpha_i(1 - p_i)$. Rearranging terms, the distortion in measured prevalence is:

$$q_i - p_i = \alpha_i(1 - p_i) - \beta_i p_i. \quad (4)$$

Here $\alpha_i(1 - p_i)$ is the share of units falsely classified as positive while $\beta_i p_i$ is the share falsely classified as negative. Our estimate of the target value is biased upward when the share of FPs, $\alpha_i(1 - p_i)$, exceeds the share of FNs, $\beta_i p_i$, and biased downward in the opposite case.

The prevalence distortion identity clarifies why accuracy ($\Pr[Z_{ij} = T_{ij}]$), while being an important characteristic of a measurement system, is insufficient for capturing measurement error induced bias. Accuracy is defined as:

$$A_i = (1 - \beta_i)p_i + (1 - \alpha_i)(1 - p_i). \quad (5)$$

Suppose the prevalence of the target value being $T_{ij} = 1$ is $p_i = 0.10$. And suppose the measurement system has high specificity ($1 - \alpha_i = 0.95$) and high sensitivity ($1 - \beta_i = 0.90$). This gives an accuracy rate of $A_i = 0.90(0.10) + 0.95(0.90) = 0.945$. Despite this high accuracy, the prevalence of the LLM reporting $Z_{ij} = 1$ is $q_i = 0.90(0.10) + 0.05(0.90) = 0.135$. So, the LLM-generated data has an

³To keep the notation simple, we focus on a single measurement system, $m = 1$, allowing us to omit the superscript.

absolute prevalence bias of 3.5 percentage points or a relative overestimate of 35% even though the measurement system classifies 94.5% of observations correctly.

2.2 Task-Level String Outcomes

A second special case of the general measurement model is data generated as strings. Continue to let $T_{ij} \in \{0, 1\}$ be the target value, which is either present or absent, and $Z_{ij} \in \{0, 1\}$ be the model’s binary measure of the target value. Let N_{ij} be the target string (in our application a variable name) that is only defined when $T_{ij} = 1$ and let X_{ij} be the LLM-generated string defined only when $Z_{ij} = 1$. In this way the binary measurement model governs the string measurement model.⁴

Semantic Similarity. We first consider the quality of the generated string conditional on a true positive in the binary task. Define semantic similarity as:

$$S_{ij} = s(X_{ij}, N_{ij}) \in [0, 1] \quad \text{if } T_{ij} = Z_{ij} = 1, \quad (6)$$

with S_{ij} undefined otherwise and $s(n, n) = 1$. The corresponding unit-level conditional semantic dissimilarity is simply $U_{ij} = 1 - S_{ij}$ with $U_{ij} = 0$ being perfect similarity, $U_{ij} = 1$ being no semantic similarity, and U_{ij} undefined when outside the TP range.

To move from the unit-level error to a system-level measure, define the mean semantic similarity score among TP observations as:

$$\mu_i = \mathbb{E}[S_{ij} \mid T_{ij} = Z_{ij} = 1]. \quad (7)$$

As perfect measurement corresponds to a semantic-similarity score of one, mean conditional semantic dissimilarity is:

$$1 - \mu_i = \mathbb{E}[U_{ij} \mid T_{ij} = Z_{ij} = 1]. \quad (8)$$

This measure evaluates measurement error in the second-stage string-generation task conditional on the model correctly detecting that the target is present.

Semantic Fidelity. Although semantic similarity is defined only for TP cases, it is useful to have a measure for all observations with $T_{ij} = 1$. We define the concept of semantic fidelity as:

$$F_{ij} = \begin{cases} S_{ij}, & \text{if } T_{ij} = 1, Z_{ij} = 1, \\ 0, & \text{if } T_{ij} = 1, Z_{ij} = 0. \end{cases} \quad (9)$$

⁴This two-stage model is exactly how we design the prompts for the LLM. Variable names are generated conditional on an affirmative answer to a binary question about the presence or absence of the concept.

A TP receives fidelity equal to the semantic similarity between the target and generated strings. A FN receives zero fidelity because the model fails to produce a string when a target string exists. The corresponding unit-level error in semantic fidelity is:

$$1 - F_{ij} = \begin{cases} U_{ij}, & \text{if } T_{ij} = 1, ; Z_{ij} = 1, \\ 1, & \text{if } T_{ij} = 1, ; Z_{ij} = 0. \end{cases} \quad (10)$$

Thus, a FN produces maximal measurement error, while a TP produces error equal to the semantic dissimilarity between the generated and target strings.

We then define the string analog of the binary measurement error model, which depends on the TP and FN rates. Taking expectations conditional on the target value being present gives us:

$$\mathbb{E}[1 - F_{ij} | T_{ij} = 1] = \beta_i + (1 - \beta_i)(1 - \mu_i) = 1 - (1 - \beta_i)\mu_i. \quad (11)$$

The string measurement error model can be decomposed into the failure to detect (β) plus the expected semantic error ($1 - \mu$) weighted by the probability of detection ($1 - \beta_i$). Average string measurement error is thus the sum of the probability of failing to detect a target that is present and the probability weighted expected semantic error conditional on detection.

2.3 Task-Level Omnibus Loss Function

Finally, we define an omnibus measurement loss function that combines classification error and semantic error. For concept i in paper j , define the unit-level loss as:

$$L_{ij} = \begin{cases} 0, & T_{ij} = 0, Z_{ij} = 0, \\ 1, & T_{ij} = 0, Z_{ij} = 1, \\ 1, & T_{ij} = 1, Z_{ij} = 0, \\ U_{ij}, & T_{ij} = 1, Z_{ij} = 1. \end{cases} \quad (12)$$

The loss is 0 for a TN, 1 for a FP or FN, and equal to the conditional semantic error for a TP. The loss function is an unweighted measure of loss, as it treats FNs and FPs as having equal maximal loss of one.⁵

Recalling that $p_i = \Pr(T_{ij} = 1)$, the task-level omnibus loss function is:

$$\begin{aligned} \mathcal{L}_i &= \mathbb{E}[L_{ij}] \\ &= \alpha_i(1 - p_i) + \beta_i p_i + (1 - \beta_i)p_i(1 - \mu_i) \\ &= (1 - A_i) + p_i(1 - \beta_i)(1 - \mu_i). \end{aligned} \quad (13)$$

⁵A more general loss function could assign distinct costs to each type of error (e.g., ω_{FP} , ω_{FN} , ω_S).

The first term is the population share of FPs, the second is the population share of FNs, and the third is the expected semantic error among TPs weighted by the population share of TPs. The final equality makes the hierarchical structure of the models explicit. When the task requires only binary classification, the semantic term is absent and omnibus loss reduces to $1 - A_i$, which is just one minus the accuracy rate.

The omnibus loss function has several nice properties. It is bounded between zero and one; it equals zero under perfect measurement; it places binary and string errors on a common scale; and it decomposes loss into interpretable components. It also provides a direct basis for comparing measurement systems in identifying a single concept, i , for a given sample or population.

That said, it is not a perfect statistic for capturing overall measurement error. Its interpretation depends on the target prevalence, p_i ; the scale of the semantic similarity function; and the maintained choice of maximal loss to both FPs and FNs. Additionally, while α_i , β_i , and μ_i are on the same zero to one scale, their values do not mean the same things.⁶ These limitations make the model suitable for comparisons across measurement systems, m , for a given concept and not across concepts with different prevalences. More fundamentally, $\mathcal{L}_i$ measures average data-level loss. It does not record where errors occur and so does not generally determine their effect on a particular substantive estimand.

2.4 Estimand Dependence

To compare measurement systems, we reintroduce the model index m and define:

$$\mathcal{L}_i^{(m)} = \alpha_i^{(m)}(1 - p_i) + \beta_i^{(m)}p_i + p_i(1 - \beta_i^{(m)})(1 - \mu_i^{(m)}). \quad (14)$$

The target prevalence p_i is common across measurement systems, while $\alpha_i^{(m)}$, $\beta_i^{(m)}$, and $\mu_i^{(m)}$ characterize system m 's performance on concept i . To connect task-level measurement error back to the substantive estimand in Equation (1), we define system-specific measurement-induced estimand distortion as:

$$\Delta_i^{(m)} = \hat{\theta}_i^{(m)} - \theta_i. \quad (15)$$

Thus, $\mathcal{L}_i^{(m)}$ measures the average quality of the model-generated data, while $\Delta_i^{(m)}$ measures the consequence of those errors for the particular substantive estimand. Measurement systems with similar omnibus losses may produce different estimand distortions if their errors occur for different papers or in different parts of the distribution.

While the target values $\mathbf{T}_i$ are observed in a human-labeled evaluation sample, they typically remain unobserved in the full sample or population. If they were observed for every paper, it

⁶Recall that α_i and β_i are percentages while μ_i is a semantic similarity score, not a percentage of incorrect semantic mismeasurement.

would mean that a human had read and labeled every paper, making the use of an LLM in the review process superfluous. The implication is that estimand distortion can be calculated for a human-labeled evaluation sample but cannot be directly calculated for the full sample. Despite this limitation, we can evaluate the dependence of results on the choice of measurement system.

We denote the mean estimate across the M measurement systems as:

$$\bar{\theta}_i = \frac{1}{M} \sum_{m=1}^M \hat{\theta}_i^{(m)}. \quad (16)$$

Then, a compact measure of cross-system dependence is the measurement system dispersion:

$$D_i(\theta) = \left[\frac{1}{M} \sum_{m=1}^M \left(\hat{\theta}_i^{(m)} - \bar{\theta}_i \right)^2 \right]^{1/2}. \quad (17)$$

As the measurement systems are a fixed collection rather than a random sample from a population of LLMs, $D_i(\theta)$ is not a standard error or confidence interval. With that said, dispersion is a useful metric, as a larger value of $D_i(\theta)$ indicates that the estimate for concept i is more sensitive to the measurement system used to generate the data.

Together, these measures distinguish three related but separate features of LLM-generated data: (1) task-level measurement error, (2) the resulting distortion in a substantive estimand, and (3) the sensitivity of that estimand to the choice of measurement system. The framework also clarifies what must be observed to evaluate each object.⁷ Target labels are required to estimate classification and semantic error rates. Without at least some labeled references one cannot determine the type and severity of measurement error in LLM-generated data. In the absence of validation data, or when examining performance on the entire corpus, system-specific estimates can be compared across measurement systems in order to determine the degree to which LLM-generated data is a function of the measurement system. But this cross-system comparison does not communicate information about measurement-system-specific performance or the presence and nature of measurement error in a single model.

3 Corpus Construction and Labeling

Operationalization of this framework requires two linked data structures: a human-labeled reference sample, which provides the target values needed to estimate classification and semantic error, and a common corpus processed by each measurement system, which permits calculation of measurement-system-specific descriptive estimates and their dispersion. Corpus definition and label construction are thus not ancillary data-management choices. Rather, they determine both the population to which the descriptive claims apply and the benchmark against which measurement quality is

⁷Appendix A provides a practical guide to operationalizing this framework for a variety of types of target estimands.

assessed. Figure B.1 in Appendix B presents our development pipeline from corpus construction through performance validation to substantive analysis.

3.1 Corpus Construction

The first decision we make in corpus construction is the choice of source for paper citations and metadata. While there are many searchable databases of scholarly work (e.g., Google Scholar, Web of Science, JSTOR, PubMed, Scopus), all are partial collections of the scientific record. We select the OpenAlex database (Priem et al., 2022) for three reasons. First, OpenAlex is the successor to the Microsoft Academic Graph (MAG), which at retirement in 2021 contained 239 million research publications. The MAG was the most complete index of scientific publications (Sinha et al., 2015) and was frequently used for meta-analysis of research trends in economics (Jones, 2021; Josephson and Michler, 2023). OpenAlex’s database is a frequently updated continuation of the MAG but adds additional sources like Crossref, ORCID, Pubmed, arXiv, and other institutional repositories. Because of the nearly daily updating to the bibliographic catalog, any query of OpenAlex is a single snapshot in time of what is in the catalog. At the time of our API call to OpenAlex (19 February 2025) the graph contained 248 million academic works, nearly three times the number in Scopus and Web of Science (Priem et al., 2022). Second, OpenAlex is not just a search tool or database, but like the MAG, it is a catalog that disambiguates and connects scholarly work. Each catalog entry contains metadata such as title, author(s), year of publication, and DOI as well as concepts that capture what the work is about, and author and institutional relationships. This information forms a series of nodes and edges that connect the metadata from one entry to another. OpenAlex indexes 65,000 concepts allowing for greater search coverage than just querying title, abstract, and keywords. Third, OpenAlex is an open access catalog, making research based on OpenAlex entries more reproducible.

The second decision we make is to construct a corpus of literature that is as close as possible to the universe of all papers that use rainfall as an IV. This decision means that our search criteria seeks to minimize type II errors at the expense of making many type I errors. Therefore, our search criteria is overly broad, relative to a search criteria that seeks to maximize accuracy by minimizing both type I and type II errors. This has the effect of increasing recall at the expense of specificity. The practical outcome of this decision is that we expect many papers in our corpus of rainfall IV papers to not actually have a rainfall IV ($T_{ij} = 0$). The downstream implication is that the number of TNs and FPs will increase, decreasing precision. We believe this decision is justified because it moves the most difficult error (i.e., an unretrieved paper) into the corpus. A non-rainfall IV paper can be correctly classified by the LLM as a TN while a paper that never enters the corpus cannot be recovered. In terms of inference for the literature as a whole, the result of admitting more FPs is that we contaminate the corpus with papers that do not use rainfall IVs, potentially distorting characteristics of the rainfall-IV literature when the LLM classifies the paper as a FP.

The alternative would be to reduce the coverage of the corpus, which if omission are not random (e.g., older papers, papers in certain fields with different IV terminology) can distort conclusions about what the literature actually says.

The third decision we make regards the actual search terms used to query OpenAlex. We use Boolean operators to search for works that include “((Weather) AND (Instrumental Variable)) OR ((Rainfall) AND (Instrumental Variable))”. This results in more than 65,000 plausibly relevant academic works. We then filter these results for only English language works and works in OpenAlex’s pre-defined domains of “Economics”, “Econometrics”, and “Finance”. This narrows the search results to 4,191 papers, comprising what is, in our view, the potential universe of economic papers that use rainfall as an IV. With the metadata on papers generated from OpenAlex we then download all works as PDFs through a combination of web scraping and manual download.

3.2 Labeling

Because our OpenAlex search prioritized recall, the candidate corpus of 4,191 papers likely includes many papers that do not contain a rainfall IV. We selected the 439 papers ($\sim 10\%$ of the total) that have the highest OpenAlex relevance scores to form our human-labeled reference sample.⁸ The selection of papers with the highest relevance score increases the number of rainfall IV papers in the human-labeled reference sample and thus the opportunities to train and evaluate string extraction. The tradeoff is that, because the human-labeled reference data is not a random subsample, validation metrics cannot be interpreted as estimates of model performance for the candidate corpus.

Given that our primary goal is to characterize whether papers use a rainfall IV, the name of that rainfall IV, and the name of the endogenous variable that it instruments, we build a human-labeled reference set with a hierarchical structure and accompanying contextual information. Table 1 presents the extraction schema, including fields, output type, example record, and dependencies. Before formal labeling began, the principal investigators (PIs) trained two undergraduate research assistants (UGRAs) and two graduate research assistants (GRAs) to apply a common coding protocol. Training covered the study objectives, causal identification, IV estimation, and the terminology used to describe endogenous variables and instruments. Annotators reviewed worked examples and received a codebook and 40 pre-labeled reference papers. They then independently coded ten calibration papers and received oral and written feedback from the PIs before proceeding to the remaining assignments.

The annotation process proceeds as follows. For each paper, annotators first record the PDF filename, article title, and DOI. If the paper is empirical, annotators assign a label of 1 and record the dependent variable(s) used in the analysis. If the paper is not empirical, annotators assign

⁸OpenAlex computes a relevance score to rank search results. The score is based on text similarity between the search terms and the metadata attached to the academic work. This semantic similarity score is then weighted by citation count, with higher cited papers receiving greater weight.

a label of 0. In this case, and whenever a dependency condition is not satisfied, the variable field is assigned **n/a**, as are all subsequent fields in the hierarchy. Conditional on the paper being empirical, annotators assign a label of 1 or 0 if the paper has an endogeneity problem or not. If there is an endogenous variable, annotators record the name of the endogenous variable(s). The same hierarchical process is applied to whether an IV is employed and if the IV is precipitation-based. For every field that contains an extracted variable name, annotators also record the section heading and the sentence or sentences where the variable name was found.

Recording section headings and the sentences containing the variable names can provide LLMs with an opportunity to identify context clues for locating and correctly labeling the dependent, endogenous, and instrumental variable.⁹ This is particularly useful in our test case because the rainfall-IV literature is extensive, interdisciplinary, and methodologically diverse. For example, explicit mention of an “identification strategy” is relatively rare outside of economics and even in economics it is rare prior to the Credibility Revolution. Information on the instruments varies by econometric method. Identification in dynamic panel models, selection models, or Lewbel estimators are reliant on IVs, yet papers using these methods rarely use terms like “exclusion restriction,” which is a common term in papers using 2SLS.¹⁰ This diversity of terminology and context means that the task cannot be solved reliably through keyword matching alone. The model must therefore infer econometric roles from the surrounding context.

To verify the accuracy of the human reference labels, we undertake a three-stage process. In the first stage, the two UGRAs and the two PIs were randomly assigned a subset of papers so that each paper was independently coded twice. In the second stage, the papers were reassigned to the two GRAs and the two PIs and independently coded twice again. In the third stage, the PIs compared the labels across stages and collectively reviewed disagreements so as to reconcile differences. The resulting adjudicated reference labels operationalize the target values T_{ij} in our measurement-error framework.

3.3 Prompt Engineering

Prompt engineering is the practice of constructing and polishing natural language inputs for LLMs to improve the quality and relevance of their outputs. Good prompt engineering can significantly enhance output quality and reduce hallucination frequency, but what makes a “good” prompt is not always clear, particularly if one lacks quantitative metrics to evaluate model performance. Different prompting formulations can often engender entirely different outputs even when provided to the same model with the same input information. Experimental results paired with our own exploration

⁹This contextual information is part of the human-labeled reference data but not part of the information the LLM is prompted to extract.

¹⁰Moreover, many of these econometric models have more than one name: e.g., Blundell-Bond or Arellano-Bond for dynamic panels; endogenous switching, Heckman selection, or hazard selection for selection models; Lewbel or heteroskedasticity-based instruments for generated instruments.

of the subject demonstrate that models can be highly reactionary even to the most subtle alterations in wording (Leidinger et al., 2023; Mizrahi et al., 2024; Salinas and Morstatter, 2024). By eliciting desired model behaviors through prompt alteration, we can observe improvements to output quality without modifying model parameters (Sahoo et al., 2025).

Our prompt engineering follows an iterative process in which we fine-tune an independent GPT-4.1 model using a draft prompt and fine-tuning training data. After training, the model outputs its own labels for comparison with the held-out evaluation set, and we calculate performance validation metrics. The validation metrics provide information on whether errors are driven primarily by FPs or FNs. We also calculate semantic similarity scores and visually compare model-labeled strings to human-labeled strings in order to determine if model-labels are conceptually aligned with the target values. Based on this information, we revise prompt text and structure and begin the next iteration on a GPT-4.1 model independent from the first.¹¹ We had no prespecified stopping rule or target level for validation metrics and semantic similarity. Rather, we stopped iterating when successive prompt revisions produced only minor marginal improvements, with the rate of performance improvement approaching zero.

Across all three implementations, we use a common document-grounded prompting framework that contains a fixed context layer and a set of 10 queries (questions). The fixed context consists of the system role (job description), extraction rules, normalized paper text, and an assistant prefill. The queries have the same hierarchical structure used by the human annotators. This hierarchical structure efficiently extracts information by reducing unnecessary queries and token use by making downstream queries conditional on preceding outputs. The queries elicit four binary classifications and six string responses from the model.

Each query is uniformly structured to define the task (e.g., “extract the article title”), recommend places to look in each PDF (e.g., “top of first page (main heading before authors/abstract)”), specify extraction rules (e.g., “ignore footnote markers (*, †, superscripts)”), and provide an explicit desired output format (e.g., “ONE line: title text only. No quotes, labels, or extra words”). We then immediately repeat the text of the query because in the linear logic of an LLM, which reads left to right, any part (token) of a prompt cannot be informed by tokens that come after it. Thus, if a prompt is structured **context-query** the **query** is informed by the **context** but the LLM encounters the **context** without having access to the **query**. Leviathan et al. (2025) shows that sending the query twice within a prompt allows each token in the query to inform every other token in the query, resulting in substantially improved performance. The full text of the queries is in Appendix B.

¹¹Our decision to iterate prompts using the same training dataset does run the risk of overfitting through prompt selection. To minimize this risk, no evaluation data enters fine-tuning and every fine-tuned model is an independent run. However, the prompts are optimized to the evaluation set and so error rates may be higher when the implementations are deployed to read the entire corpus.

4 Model Implementations

Researchers using LLMs for information extraction are faced with a set of tradeoffs, among which are task specialization, output quality, and the required data, computation, and labor. An off-the-shelf model can be used with task-specific prompts alone, supplied with reference examples through retrieval, or adapted through additional training. We compare these approaches using the same general-purpose base models. The implementations differ in whether human-labeled reference information enters the prompt or is used to update model parameters.

We use GPT-4.1 through OpenAI’s API to compare three different implementations: (1) a simple zero-shot prompting strategy, (2) retrieval-augmented generation (RAG), and (3) supervised fine-tuning (SFT). Each paper is processed in a dedicated thread that becomes a single multi-turn conversation record containing the fixed context, the sequence of query/response turns for each extraction field. This allows later queries to draw on the model’s response to earlier queries within the same conversation. In the final step, model output passes through some post-hoc consistency processing.

We randomly divide the paper-level records into a non-held-out reference set (80%) and a common held-out evaluation set (20%). The non-held-out reference set supplies retrieval examples for RAG and training and development data for SFT. The common held-out evaluation set supplies the targets against which generated outputs from all three implementations are compared. The labels from this common held-out set are neither retrieved as examples nor used to update model parameters.

4.1 Zero-Shot Baseline

Our zero-shot implementation is a document-grounded, “closed-book” prompting strategy. During extraction, the model receives the normalized text of the focal paper and the prompts described in Section 3.3, but has no access to the human-labeled reference data. Zero-shot therefore isolates what the model can extract from the focal paper using the prompting framework alone and provides the baseline for evaluating the additional value of RAG and SFT.

Figure 1 lays out the zero-shot workflow. Each prompt supplied to the model contains a fixed context layer, the sequence of queries, and a set of dependency rules and consistency checks. Panel 1 shows the fixed context supplied to the model for each paper which consists of the system role, user rules, normalized paper text, and assistant prefill. The assistant primes the model’s behavior with a fake commitment with the intention of steering later responses towards our preferred behavior. This is a would-be-expected response from a model that was fed the same system context and user queries.

Panel 2 shows the sequential query/response loop. The primed model is fed each of our specified queries one at a time with the text of the current query repeated twice. After the first call, the model

also receives all prior query/response turns. This creates a multi-turn sequential query/response loop, where the conversation history accumulates as the model moves through the sequence of queries.

Panel 3 shows how the dependency structure governs which queries are sent to the model. Before each dependent query, the script checks the relevant response from the earlier query. If the dependency is not satisfied, the query and any downstream dependent queries are skipped, and the corresponding fields are coded as **n/a** without calling the model. After all eligible queries have been processed, a post-hoc consistency pass re-checks the four dependency gates and enforces field dependencies in the ten-field record. As zero-shot uses neither retrieved examples nor parameter updates, it is our least information-intensive implementation.

4.2 Retrieval Augmented Generation (RAG)

RAG is a more information-intensive implementation of the same prompting strategy used in zero-shot. The focal paper, fixed context, field-specific queries, dependency structure, and creation of conversation history remain unchanged. What differs is that each eligible query is augmented with examples retrieved from the human-labeled reference data. RAG is therefore an “open-book” implementation. The model has access to adjudicated reference examples through the prompt, but its model weights remain unchanged (Sahoo et al., 2025).

Figure 2 shows the two phases of our RAG implementation. Panel 1 covers indexing, while Panels 2 and 3 show the retrieval-conditioned generation loop. During indexing, we use the non-held-out reference set to construct five field-specific retrieval pools as vector indices. The first pool contains general information used for title and DOI extraction. The remaining four contain information on dependent variables, endogenous variables, instrumental variables, and rainfall IVs. Each reference entry contains an article excerpt, its embedding, and the corresponding adjudicated answer. The resulting indices remain fixed when the models are applied to the held-out evaluation set and the candidate corpus.

Panel 2 shows the retrieval process for each eligible query. The pipeline extracts keyword-specific passages from the focal paper using a fixed set of terms related to our query concepts. The extracted focal-paper context is embedded and compared with reference-entry embeddings in the corresponding retrieval pool. To ensure we are injecting context relevant to the focal paper, we calculate cosine similarity scores between each query and all pool embeddings. Let ϕ_{ij} denote the embedding of the focal-paper context for concept i in paper j and $\psi_{\ell,k(i)}$ the embedding of reference entry ℓ in retrieval pool $k(i)$. We calculate cosine similarity as

$$\text{sim}(\phi_{ij}, \psi_{\ell,k(i)}) = \frac{\phi'_{ij} \psi_{\ell,k(i)}}{\|\phi_{ij}\| \|\psi_{\ell,k(i)}\|}. \quad (18)$$

The three most similar reference entries are retrieved for each query. Each is supplied to the model

as an article excerpt paired with its adjudicated answer. Panel 3 shows how these examples are combined with the fixed context, current query, and prior query/response history to form the model prompt. After each call, the query/response pair is appended to the conversation history. If eligible queries remain, the loop returns to Panel 2 to retrieve examples for the next query. Once all eligible queries have been processed, the ten-field record undergoes the same post-hoc consistency checks as in zero-shot.

The principal advantage of RAG in our setting is that it gives an otherwise unchanged model access to examples of how human annotators interpret heterogeneous economic terminology. Conditioning output generation on retrieved context can improve factual grounding and reduce hallucinations (Lewis et al., 2020). But RAG also introduces an additional potential source of error. Performance depends not only on the model but on the quality of the reference data and whether the retrieval procedure identifies substantively relevant examples. Semantically similar text need not represent the same economic concept. Increasing the size of the retrieved context window raises the probability of surfacing substantively relevant examples, but can also introduce less relevant material that dilutes the contribution of the most informative examples. More context thus does not guarantee better performance (Gu, 2026).

4.3 Supervised Fine-Tuning (SFT)

SFT uses the human-labeled reference data to adapt model parameters before extraction. Unlike RAG, which supplies examples within each prompt, SFT uses adjudicated responses to modify how the model responds to subsequent queries. In our setting, the aim is to improve interpretation of the indirect phrasing, qualifications, and domain-specific terminology used to describe endogeneity and IV strategies. We fine-tune the same base models used in zero-shot and RAG. The tuning mechanism is represented by Low-Rank Adaptation (LoRA), a parameter-efficient fine-tuning method (Hu et al., 2021).¹²

Figure 3 separates fine-tuning from subsequent extraction. Panel 1 shows the allocation of reference data and the training process. We divide the non-held-out reference set 7:1 into a training and development set. The development set is called the validation set by fine-tuning platforms. The pretrained model enters fine-tuning with its base weights fixed.¹³ For each selected weight matrix $\mathbf{W}$, LoRA introduces two smaller trainable matrices, $\mathbf{A}$ and $\mathbf{B}$. Their product provides a low-rank adjustment to the original weights such that:

$$\mathbf{W}' = \mathbf{W} + \mathbf{BA}. \quad (19)$$

¹²OpenAI does not disclose its parameter-updating method but the generative AI community is in general agreement that GPT fine tuning uses LoRA as all other major fine tuning platforms use LoRA. That said, LoRA that GPT uses LoRA is an assumption not a verified description of the platform’s implementation.

¹³This differs from full fine-tuning, in which elements in a series of matrices that make up a base model’s architecture (e.g., receiving, embedding, predicting, etc.) are made adjustable, whereby the underlying model is fundamentally modified to fit a specific task.

Only the adapter matrices are updated during training. This reduces the number of trainable parameters relative to full fine-tuning while allowing the model’s responses to adapt to the extraction task.

The training loop in Panel 1 processes batches of labeled conversation records. At each step, the model computes token probabilities for the adjudicated responses and the cross-entropy loss associated with those responses. The loss is used to update $\mathbf{A}$ and $\mathbf{B}$, while $\mathbf{W}$ remains fixed. The updated adapter matrices are used in the next training step, and the process repeats across batches and epochs. Within the training records, later queries are conditioned on the adjudicated responses to earlier queries. During extraction, by contrast, later queries are conditioned on the model’s own prior responses. The fine-tuning development set is evaluated during training and at completion. Development-set metrics allow us to monitor performance beyond the training examples without using those records for parameter updates. We deploy the final model returned by the completed job rather than selecting an earlier checkpoint based on development-set performance.

Panel 2 shows extraction using the final fine-tuned model. Each paper begins a new thread with the same fixed context and field-specific queries used in zero-shot. Eligible queries are processed sequentially, and the same dependency rules and post-hoc consistency checks produce a ten-field record for each paper. No reference examples are retrieved and no parameters are updated during extraction. Relative to zero-shot and RAG, SFT changes the model used to process the prompt rather than the information supplied about the focal paper or relevant examples. Despite requiring a higher upfront investment in computation and labor, SFT can scale more efficiently than zero-shot or RAG approaches (Liu et al., 2024; Dunn and Dagdelen, 2022).

5 Performance Validation

Prior to deploying each implementation on the entire corpus, we evaluate performance based on a set of metrics designed to capture accuracy and to quantify the types of successes and failures.

5.1 Validation Criteria

We evaluate binary classification and string extraction performance using the validation metrics defined in Section 2. Following a modified version of the answer evaluation rubric in Balaguer et al. (2024), we also conduct a targeted manual review of model-implementation outputs to characterize common failure modes and ensure that apparent quantitative gains correspond to substantively correct extraction.

Confusion matrices summarize agreement between the LLM-generated measures (Z_{ij}) and target values (T_{ij}) by cross-tabulating the target values (rows) against the measured values (columns). Each confusion matrix contains four cells: true positives (TP), cases where $T_{ij} = Z_{ij} = 1$; true negatives (TN), $T_{ij} = Z_{ij} = 0$; false positives (FP), cases where $T_{ij} = 0$ and $Z_{ij} = 1$; and false

negatives (FN), cases where $T_{ij} = 1$ and $Z_{ij} = 0$. Table 2 visualizes these quantities. Together, they characterize the measurement system’s error profile by distinguishing false detections from missed positive cases while also recording correct detections and correct rejections.

From the confusion matrix cells we report accuracy, sensitivity, specificity, precision, F1-score, and balanced accuracy. Accuracy is the share of correct classifications: $(TP + TN) / (TP + TN + FP + FN)$. Sensitivity (the true positive rate) is calculated as $TP / (TP + FN)$ and measures how often target-positive cases are detected. Specificity (the true negative rate) is calculated as $TN / (TN + FP)$ and measures how often target-negative cases are correctly rejected. Precision (the positive predictive value) is calculated as $TP / (TP + FP)$ and captures the reliability of positive predictions. The F1-score is the harmonic mean of precision and sensitivity and summarizes positive-class performance when both FPs and FNs matter. Balanced accuracy is the arithmetic mean of sensitivity and specificity and is particularly informative when class frequencies are uneven, because it weights performance on the positive and negative classes equally.

For non-binary responses, we evaluate semantic agreement between the model-generated string (X_{ij}) and the target string (N_{ij}) using sentence embeddings and cosine similarity. We encode each string using all-MiniLM-L6-v2, a SentenceTransformer model built on a Bidirectional Encoder Representations from Transformers (BERT) family architecture (Reimers and Gurevych, 2019). SentenceTransformers are transformer-based models that are fine-tuned to produce sentence-level embeddings for semantic similarity. For TP cases, semantic similarity is $S_{ij} = s(X_{ij}, N_{ij})$ where s is the cosine similarity between the two embeddings. This allows us to measure semantic alignment beyond exact string matching and to summarize performance using both the mean and distribution of semantic similarity across the evaluation set.

5.2 Performance Validation

Figure 4 summarizes task-level evaluation metrics on the common held-out evaluation set for the three implementations of GPT-4.1. Panel A presents the confusion mosaics for each binary outcome. Panel B uses the raw confusion matrices to calculate binary task-level validation metrics. Panel C presents range, mean, and standard deviation of semantic similarity scores while Panel D draws the distribution of these scores.

What is immediately obvious is that all implementations perform very well on binary classification tasks or classification tasks. Of the 88 papers in our held-out evaluation sample, each implementation misclassifies (FP or FN) 12 or fewer papers for every binary query. Consequently, all implementations have high accuracy (share of correct answers) and precision (positive predicted value). In fact, accuracy for the three implementations on the empirical analysis, IV use, and rainfall IV tasks varies by only three to four percentage points. The difference in accuracy is larger for endogeneity, driven by RAG’s relatively poor performance, but even on this task the largest difference is only eight percentage points.

Despite this high accuracy and precision, when implementations make errors the type of errors differ substantially. Measures of specificity (the TN rate) and sensitivity (the TP rate) capture these differences. The RAG implementation produces noticeably more FPs than either zero-shot or SFT for empirical analysis, endogeneity, and IV use. These errors reduce specificity and balanced accuracy. As an example, for empirical analysis RAG achieves accuracy of 90% but specificity of only 36%. In other words, RAG is misidentifying papers as being empirical, having an endogeneity problem, and using an IV at a higher rate than zero-shot or SFT.

The zero-shot implementation makes almost as many errors as RAG but exhibits a different error profile. It has very few FPs but many more FNs for endogeneity, IV, and rainfall IV, resulting in higher specificity but lower sensitivity. Thus, zero-shot is missing papers that actually do have an endogeneity problem, use an IV, and use a rainfall IV. Empirical analysis is the exception where zero-shot has sensitivity levels that match those of the other implementations. In fact, all three implementations correctly identify all 74 papers that are empirical, yielding perfect sensitivity, with performance differences arising entirely from FPs among the remaining 14 papers.

SFT generally has the strongest and most balanced performance of the three implementations. It produces relatively few FPs and FNs across all four tasks and performs particularly well in detecting the presence of a rainfall IV, where it only misses one positive and two negative cases. Its performance is nearly as strong for IV use and endogeneity problem, with only a small number of errors of either type. As a result, SFT combines sensitivity and specificity more consistently than RAG and avoids the stronger FN tendency of zero-shot. This produces high balanced accuracy in addition to high accuracy, precision, and F1-scores. Taken together, the results reinforce why accuracy alone is insufficient for evaluating LLM-generated data. While all three implementations are highly accurate, there are meaningful differences in when and how implementations get it wrong, and these different patterns of measurement error can translate into downstream estimand distortion.

Panels C and D show a different error pattern for string extraction tasks. Semantic similarity is strongest when the string extraction tasks is relatively simple metadata, like title (≈ 0.97) and DOI (> 0.87). Performance declines and diverges across implementation when the task requires greater contextual interpretation to identify substantive economic concepts. Similarity scores for dependent variables, endogenous variables, instruments, and rainfall instruments are markedly lower and considerably more variable. The zero-shot and RAG implementations perform similarly on these more demanding fields, with mean semantic similarity generally between 0.50 and 0.70. SFT consistently produces higher similarity scores, with means roughly between 0.65 and 0.75. The largest differences in semantic similarity arise in dependent-variable extraction. Zero-shot and RAG scores spread broadly across the scale, with considerable density near zero, while SFT places more density on high-scoring outputs. Mean similarity is 0.73 for SFT, compared with 0.49 for zero-shot and 0.47 for RAG.

For the other variables names, SFT always outperforms zero-shot and RAG in terms of mean score, though sometimes these difference can be very minor. For instrument names, all three produce nearly identical means: zero-shot (0.70), RAG (0.69), and SFT (0.70). Nevertheless, the similarity in means masks considerable variation in the quality of individual extractions. SFT concentrates more density at high scores but its advantage is not uniform. RAG has higher means than zero-shot, yet typically has larger mass near zero, meaning RAG produce some very poorly matched strings. These comparisons cover only evaluated strings and do not necessarily include the same papers across implementations. They therefore describe the quality of recovered strings rather than whether every required string was recovered. Higher average similarity does not establish consistently accurate or complete extraction.

Taken together, the panels distinguish successful classification from accurate extraction. An implementation can correctly detect that a concept is present while producing a poorly matched description of that concept. Neither high classification accuracy nor a high mean similarity score fully characterizes measurement error in the generated data.

6 Results

We next examine whether differences in task-level measurement performance translate into substantive differences in the description of the rainfall-IV literature. As we do not have target labels for the candidate corpus, we cannot directly calculate estimand distortion. Instead, we compare the labels generated by the three implementations paper-by-paper to characterize how strongly measured facts depend on implementation. Then we use each dataset separately to construct Sankey diagrams that map the relationship between rainfall IVs and the endogenous variables that are being instrumented. By visualizing these nodes and their connecting edges, we can see how economists use rainfall IVs and determine the extent to which our conclusions about the literature are a function of the measurement system used to summarize the literature.

6.1 Paper-Level Comparison

Each implementation outputs a data set in which the unit record is a single paper and the generated labels are the responses to the queries in the prompt. We focus on rainfall identification because it determines which papers enter the substantive analysis. As can be seen in Table 3, of the 3,840 papers in the common model-labeled corpus, the zero-shot implementation identifies 160 papers as using a rainfall IV while the RAG identifies 168 papers. SFT identifies 187 papers, or about 11% more than RAG and 17% more than zero-shot. The union of these sets contains 218 papers that are identified as using a rainfall IV by at least one implementation. Of these, all three implementations agree on 137 papers (63%), exactly two agree on 23 papers (11%), and 58 papers are identified by exactly one implementation (27%). Thus, rainfall-IV status differs across measurement systems for

37% of papers identified by at least one system. For a more precise metric of measurement system dependence, we calculate the pairwise Jaccard index, which is the intersection of two sets divided by their union. For zero-shot and RAG, the Jaccard overlap is 0.79, for zero-shot and SFT the overlap is 0.70, and for RAG and SFT the overlap is 0.70. Thus there is meaningful measurement-system dependence, at least on the extensive margin. Anywhere from one fifth up to nearly a third of the data generated by one implementation differs from the data generated by another implementation.

Conditional on the 137 papers that all three implementations agree on as having a rainfall IV, there is still substantial disagreement on what the actual rainfall IV is and what it instruments. After a basic normalization, all three implementation gives exactly the same endogenous string for 37 papers (27%), the same rainfall IV string for 24 papers (18%), and exactly the same rainfall-IV/endogenous-variable pair for just 8 papers (6%). These numbers are an upper bound on disagreement because we are enforcing exact string matches. While a substantial minority of the 137 papers are in substantive semantic agreement, there are still real and meaningful disagreements between implementations. For example, the same paper is characterized as instrumenting rye prices/real wages by zero-shot and RAG but as instrumenting poverty by SFT. Another is characterized as instrumenting for output by zero-shot and RAG but as instrumenting for a price index by SFT. There are also papers in which one implementation identifies as having an endogenous variable while another implementation identifies as n/a. These type of disagreements actually change an edge in the causal network not merely a change in the label of a node.

6.2 Relationship-Level Comparison

In developing Sankey diagrams for each implementation, we minimize redundancies in model outputs by systematically cleaning and clustering identified rainfall IVs and endogenous variables. From the common model-labeled corpus, we extract each cell entry as a base string, normalize and separate entries containing multiple variables, and send the resulting sub-strings to GPT 5 (with high reasoning) for categorization. Highly similar clusters are automatically merged and then passed through a second LLM call that consolidates them into broader categories. The resulting clusters form the two sides of the Sankey diagrams, with bands representing instrumenting relationships and band width indicating frequency. The Sankeys therefore allow us to move beyond the paper-level comparison of measurement-system dependence and ask whether those differences change the substantive inference we draw from the literature. Following Mellon (2025), the relevant feature is not simply how often particular rainfall measures or endogenous variables appear, but the relationships between them. Evidence that the same or closely related rainfall measures predict multiple endogenous variables identifies potential alternative causal pathways through which rainfall may affect socioeconomic outcomes and therefore raises concerns about the exclusion restriction in rainfall IV applications.

The zero-shot implementation identifies 173 rainfall-IV/endogenous-variable relationships grouped

into 21 rainfall IV categories and 29 endogenous variable categories. General Rainfall/Precipitation Level (30), Rainfall/Precipitation Anomalies (11), and Rainy Days & Spells (4) and all connect to Agriculture: Production & Yields (11), Crime & Security (1), and Macroeconomic Growth & GDP (21). Thus, distinct measures of rainfall (levels, deviations, counts) are used to instrument the same types of endogenous variables. Looking at the graph from the other direction, Agriculture: Production & Yields receives links from three different measures of rainfall, Crime & Security from two, and Macroeconomic Growth & GDP from six. Here, a single socioeconomic measure is correlated with numerous different ways to quantify rainfall.

RAG identifies 187 rainfall-IV/endogenous-variable relationships, grouped into 20 rainfall categories and 25 endogenous categories. Total/General Rainfall Level (51) is substantially more common than its zero-shot counterpart, while Deviations & Anomalies (10) and Rainy Days & Spells (5) are essentially unchanged. Nevertheless, all three connect to Agriculture & Rural (21), Conflict, Crime & Unrest (8), and Economic Growth & Macro-Performance (31) track the same broad set of processes. Similar to zero-shot, many different rainfall metrics feed into a single endogenous variable. Agriculture & Rural receives links from nine different measures of rainfall, Conflict, Crime & Unrest from six, and Economic Growth & Macro-Performance from seven. Despite these compositional differences in what the actual variables are and their relationships to each other, the broad network structure remains similar to zero-shot.

SFT identifies 187 rainfall-IV/endogenous-variable relationships, grouped into 21 rainfall IV categories and 22 endogenous variable categories. Rain Amount & Units (42) remains prominent, while Deviations/Anomalies/Shocks (27) increases substantially relative to the other two implementations, and Rain Days & Frequency (3) remains similar across all implements. Though qualitatively different, all three measures of rainfall again connect to Agriculture & Rural Economy (18), Conflict, Crime & Unrest (9) change little relative to RAG but SFT finds many fewer endogenous macroeconomic indicators than RAG. SFT therefore alters the relative importance of particular nodes and relationships while preserving the same network structure as produced by zero-shot and RAG.

Read from left to right, the diagrams identify potential alternative pathways that rainfall IV applications need to address. For example, a plausible causal pathway exists that connects agriculture production to income to conflict to macroeconomic performance. Yet rainfall instruments for all four. A study that uses a rainfall shock to instrument agricultural production in a regression of conflict must justify why rainfall does not affect conflict through income independently of agricultural production. Especially when a different study uses a rainfall shock to instrument income in a regression of conflict. The more variables that rainfall instruments for along a plausible causal pathway, the harder it is to maintain the assumptions required for the exclusion restriction to hold in any particular application. The Sankeys make this concern visible. The relevance argument for one endogenous variable identifies a competing pathway for another application.

Read from right to left, the same diagrams raise a different concern. Agricultural variables receive links from annual rainfall levels, seasonal levels, anomalies, variability, and the frequency, instead of the volume, of rainfall. Macroeconomic and household-welfare variables likewise receive numerous different types of rainfall instruments. Differences in setting and theory may justify using one or more measures of rainfall to instrument for a specific endogenous variable. But these measures cannot simply be assumed to be interchangeable. The range of alternatives used in the literature creates scope for choosing from the many plausible instruments the rainfall metric that produces the desired first-stage results. The rainfall IV we observe may be the one that was selected not because of its theoretical fit or economic rationale but for its significance.

The Sankey diagrams reveals measurement-system dispersion in the composition and relative importance of individual rainfall-instrument relationships, but considerably less dispersion in the broader conclusions we draw from the literature. Across implementations, the network retains the same basic structure. Many measures of rainfall are used to instrument a wide range of distinct economic and social processes. Thus, while measurement-system choice affects which relationships appear and how prominent they are, it does not materially change the substantive concerns raised by the literature. The overall variety of relationships, rather than implementation differences in any one relationship, raises concerns about exclusion restriction violations and selection on significance.

7 Discussion

The empirical results confirm the implications of our theoretical framework that measurement system dispersion, $D_i(\theta)$, is a function of the estimand rather than being an intrinsic property of the generated data.

The decline in measurement system dispersion as the estimand coarsens highlights an important tension in the use of LLMs for literature reviews, and in the research process more broadly. Measurement-system choice matters little when one wants to establish broad descriptions of a literature. Yet this is where LLMs add relatively little. A competent traditional literature review can already establish conclusions like “the rainfall-IV literature has an exclusion restriction problem,” “trade liberalization tends to raise income,” or “climate shocks affect agricultural production.” The comparative advantage of LLMs lies instead in making fine-grained characterization at the scale of an entire literature. But this is precisely the setting where measurement error is most pronounced and measurement-system dispersion is largest.

Mellon (2025) provides a useful benchmark for this tradeoff between scale and measurement-system dependence. He reads 289 papers, manually documents relationships between weather and socioeconomic variables, using those paper-level relationships to argue that weather instruments face widespread potential exclusion restriction violations. We recover the same broad conclusion regardless of which measurement system generates our data. But the fine-grained evidence differs. Mellon’s (2025) individual relationships come from direct expert human review. Our exercise

expands the review to a candidate corpus of 4,191 papers, but the papers included, the rainfall IVs, endogenous variables, and the relationship between them all vary across measurement system. The LLMs therefore reliably reproduce the conclusion that a human review has already established while becoming most susceptible to measurement error at the level of detail that constitutes their principal value added.

The importance of estimand resolution also changes what performance validation metrics tell us in applications that use LLMs to generate data from the scientific literature. Briggs et al. (2026) provide a useful example. They show that carefully validated LLMs can recover structured information from nearly 100,000 academic papers at a fraction of the cost of human coders. They report accuracy, precision, sensitivity, and specificity typically ranging from 0.75 to 1.00. These metrics establish that the generated data closely reproduce their validation labels, but they do not establish how residual measurement error affects a particular estimand constructed from those data. The principal results in Briggs et al. (2026) are highly aggregated across a very large corpus, a setting in which residual paper-level disagreement likely has relatively little effect on the resulting conclusions. The same degree of residual error may be much more consequential when the estimand concerns an individual paper, concept, or relationship. Because Briggs et al. (2026) select a preferred model and do not carry alternative measurement systems through the analysis, their validation exercise cannot reveal how sensitive their substantive results are to model choice. High task-level performance is not, by itself, evidence that a substantive estimand is robust to the choice of measurement system.

Garg and Fetzer (2025) provide a complementary example. They use LLM-generated descriptions of causes, effects, and exogenous variation to construct fine-grained claim graphs for 44,852 academic papers, which they then use to characterize causal structure and novelty in the economics literature. Their validation results are substantially stronger for coarse method classification than for the semantic content from which claim-graph edges are constructed. They report accuracy, precision, sensitivity, and specificity with values typically ranging from 0.51 to 0.94, while mean semantic similarity with expert-labeled causes, effects, and sources of exogenous variation ranges from only 0.14 to 0.27. These metrics, together with repeated extraction, snippet-based checks, external benchmarking, and sensitivity to alternative aggregation thresholds, establish stability and agreement with external labels along important dimensions of their chosen extraction system, but they do not establish how residual measurement error affects a particular estimand constructed from those data. Unlike the highly aggregated setting in Briggs et al. (2026), the distinctive contribution of Garg and Fetzer (2025) depends directly on fine-grained paper-level measurement, precisely the level of estimand resolution at which both their semantic-validation results indicate considerable error and our results show measurement-system dependence to be most acute. Further, as Garg and Fetzer (2025) do not carry alternative measurement systems through the analysis, their robustness exercises cannot establish whether the paper-level claim graphs, or the conclusions

drawn from those graphs, would persist under a different measurement system. Their results thus illustrate that even extensive task-level validation and robustness checks within a chosen extraction system do not, by themselves, establish the robustness of downstream substantive estimands to measurement-system choice.

8 Conclusions

Literature reviews are exercises in measurement. Researchers define a population of papers, read them, code information, and use those data to characterize what a literature says. LLMs dramatically increase the number of papers and the level of detail that can be coded. But the data they produce are still measurements. When those measurements are imperfect, the relevant question is not simply how often the model succeeds or fails, it is whether the errors change what a researcher concludes.

In this paper, we develop a measurement error framework for LLM-generated data and apply it to a review of how economists use rainfall as an instrumental variable. We start with a model of measurement error that distinguishes between types of tasks as well as types of errors. The model highlights that standard validation metrics reveal little about how task-level measurement errors distort inference on substantive estimands further downstream. We then compare zero-shot, RAG, and SFT implementations of GPT-4.1. We validate each against a common held-out validation set and use each measurement system to independently construct the data used in the review. This allows us to compare task-level performance with the stability of the resulting substantive conclusions.

Our results are encouraging for simple tasks but less so as we demand more from the model in terms of contextualization and interpretation. All three implementations perform well when asked to classify whether a paper is empirical, has an endogeneity problem, uses an IV, or uses rainfall as an IV. Performance is more heterogeneous when we ask the implementations to identify and name the economic concepts underlying those classifications. SFT generally improves on these more difficult extraction tasks relative to zero-shot and RAG, but substantial semantic errors remain across all three measurement systems. Applied across the common model-labeled corpus, the implementations frequently disagree on actual variables names, what the relationship is between IV and endogenous variable, and even which papers are part of the rainfall-IV literature. High task-level performance therefore does not imply that the datasets produced by different measurement systems are interchangeable.

Yet these differences in the generated data do not always change what we infer about the literature. Across all three implementations, the same broad picture of the rainfall-IV literature emerges. Rainfall instruments a wide range of socioeconomic processes, while the same types of endogenous variables are instrumented using many different measures of rainfall. Read in one direction, these relationships identify potential alternative causal pathways that threaten the exclusion restriction.

Read in the other direction, they reveal substantial researcher degrees of freedom over how to quantify rainfall, creating scope for selection on significance. These broad conclusions survive the choice of measurement system but could also be produced by a human review of a much smaller number of papers. The finer details of the literature, which is where LLMs appear to have a comparative advantage, do not.

Selecting a preferred measurement system based on accuracy, precision, sensitivity, specificity, or semantic similarity is not sufficient to ensure credible inference. These metrics tell us how well a measurement system reproduces human labels on a evaluation sample. They do not tell us whether the remaining measurement error matters for the inference those data are meant to support.

Taken together, our results suggest a simple rule: validation should match the resolution of the claim. Absent that, researchers should demonstrate that inference on substantive estimands are robust to the choice of measurement system. When LLM-generated data are used only to characterize a literature in broad terms, measurement system choice may have little effect on the conclusion. But broad conclusions are where LLMs offer the least advantage over traditional literature reviews. The comparative advantage of LLMs lies in producing fine-grained information across a literature at a scale human readers cannot feasibly achieve. But that advantage comes with a risk. The finer the estimand, the more inference can depend on the measurement system that generated the data. The promise of LLMs is therefore not simply that they allow researchers to measure more. It is that they allow researchers to make more detailed claims from those measurements. The credibility of those claims requires showing that substantive inference survives the choice of measurement system.

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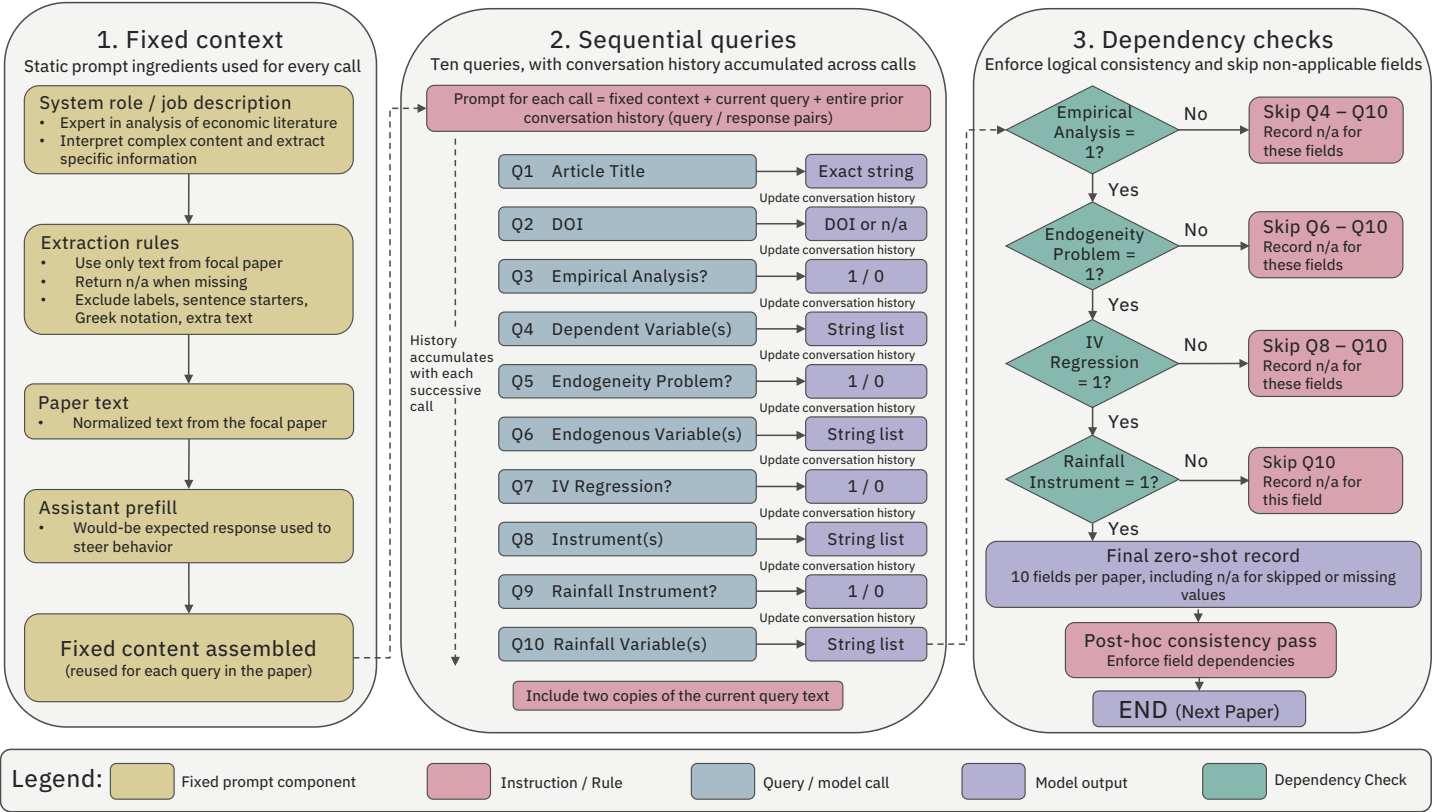
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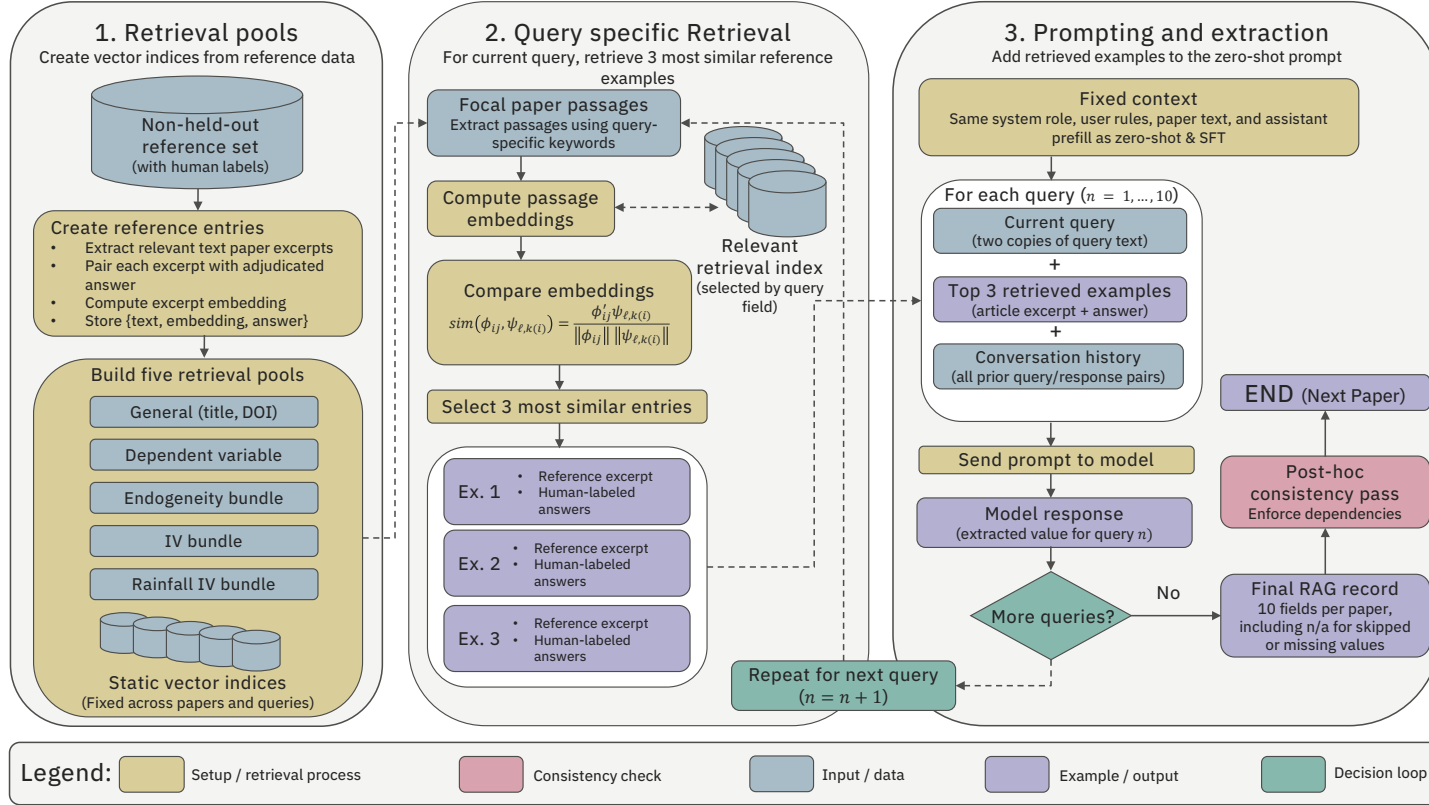
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Figure 1: Zero-Shot Implementation - Prompt Contents, Query Sequence, and Outputs



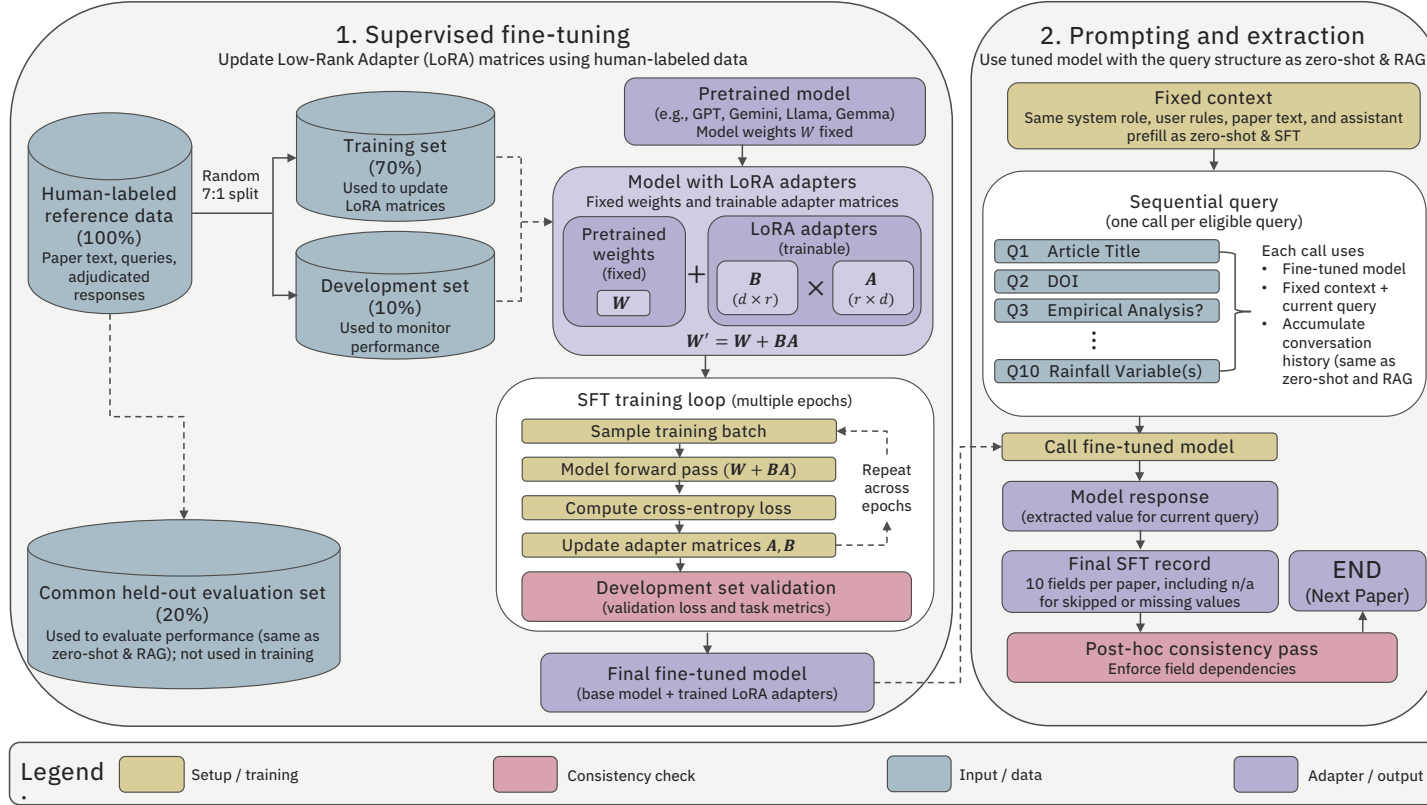
Note: The figure shows the document-grounded, closed-book zero-shot workflow. The model receives the focal paper but no human-labeled reference examples. Panel 1 assembles the fixed context from the system role, extraction rules, normalized paper text, and assistant prefill. Panel 2 processes eligible queries sequentially. Each call includes the fixed context, two copies of the current query, and all prior query/response pairs. Panel 3 shows the dependency checks that determine which queries are skipped and the post-hoc check that enforces field dependencies. Each paper yields a ten-field record, with n/a for skipped or missing values.

Figure 2: RAG Implementation - Reference Indexing, Retrieval, and Outputs



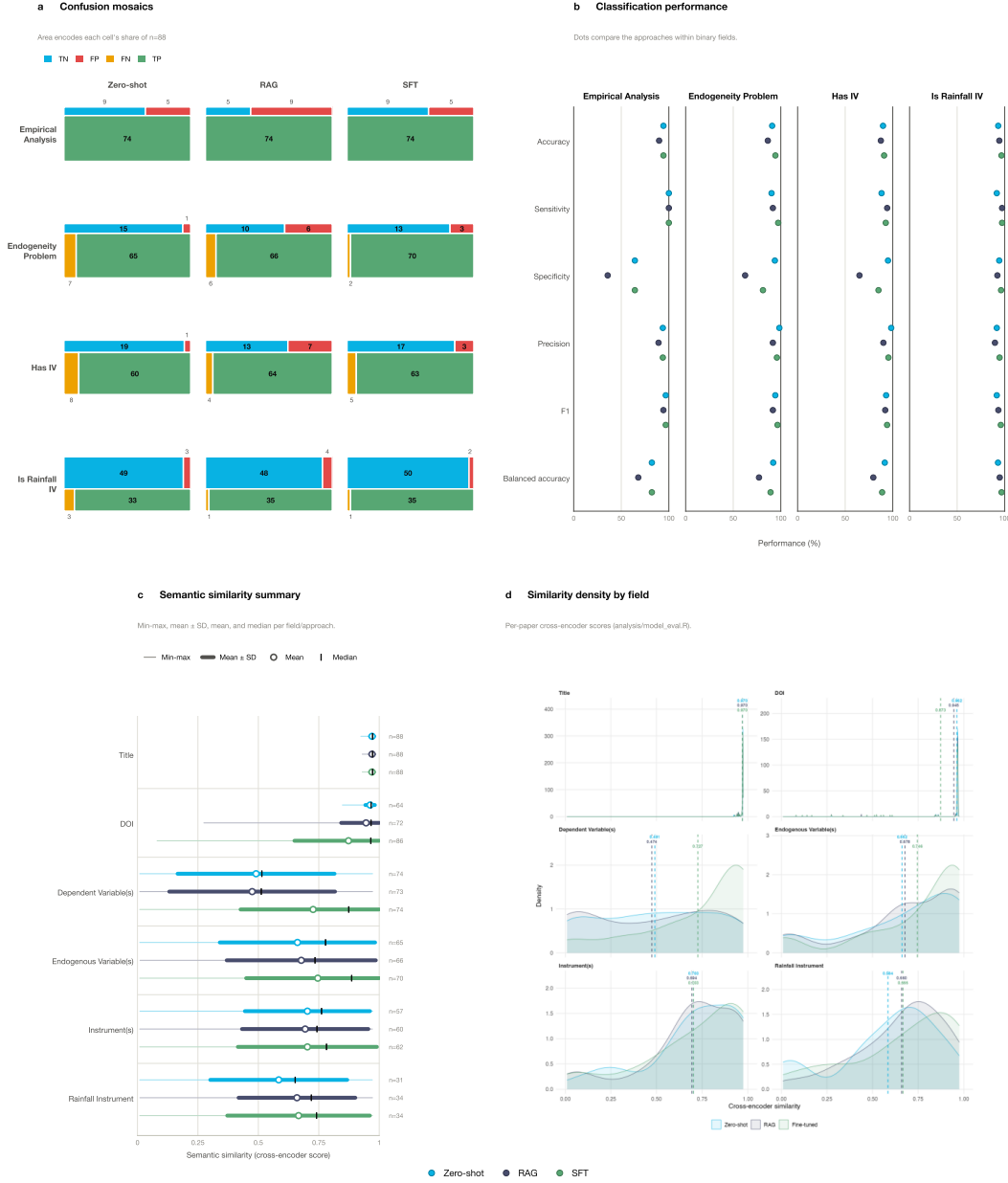
Note: The figure shows the document-grounded, open-book RAG workflow. Panel 1 constructs five static retrieval pools from the non-held-out human-labeled reference set. Each entry contains an excerpt, its embedding, and an adjudicated answer. Panel 2 embeds query-specific focal-paper passages and uses cosine similarity to select the three most similar entries from the relevant pool. Panel 3 combines these examples with the fixed context, current query, and prior query/response pairs. Retrieval repeats for each eligible query, while model parameters remain unchanged. A post-hoc check enforces field dependencies. Each paper yields a ten-field record, with n/a for skipped or missing values.

Figure 3: SFT Implementation - Training, Prompting, and Outputs



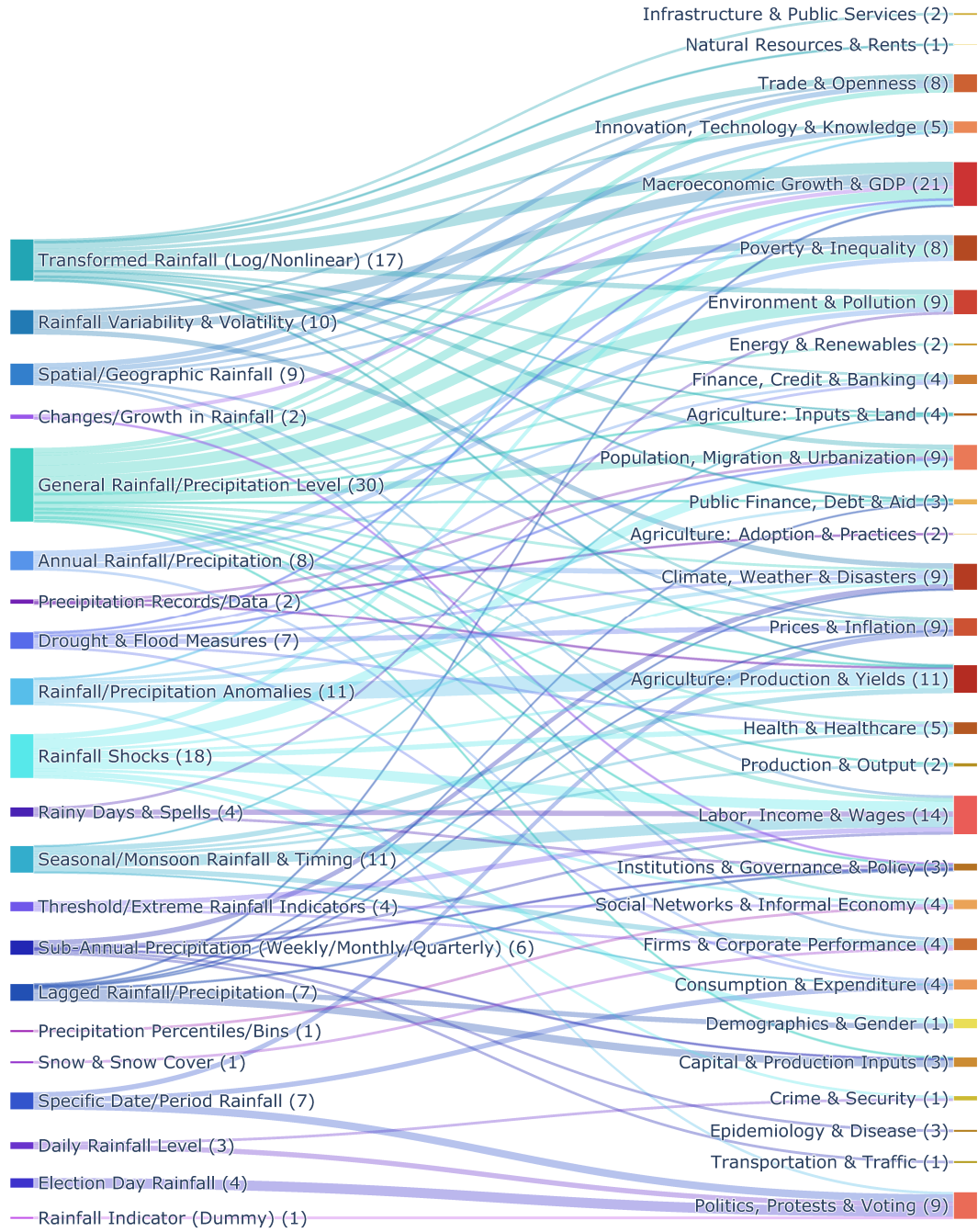
Note: The figure shows supervised fine-tuning followed by paper-level extraction. Panel 1 partitions the human-labeled reference sample into training (70%), development (10%), and common held-out evaluation (20%) sets. Training minimizes cross-entropy loss by updating low-rank adaptation (LoRA) matrices A and B , while pretrained weights W remain fixed. The development set monitors performance without updating parameters. The held-out set is used for performance evaluation, not training. Panel 2 applies the final fine-tuned model to each paper using the fixed context, current query, and accumulated query/response history. Extraction uses no retrieved examples or further parameter updates. A post-hoc check enforces field dependencies in the ten-field record, with n/a for skipped or missing values.

Figure 4: GPT: Evidence, Performance, and Semantic Stability



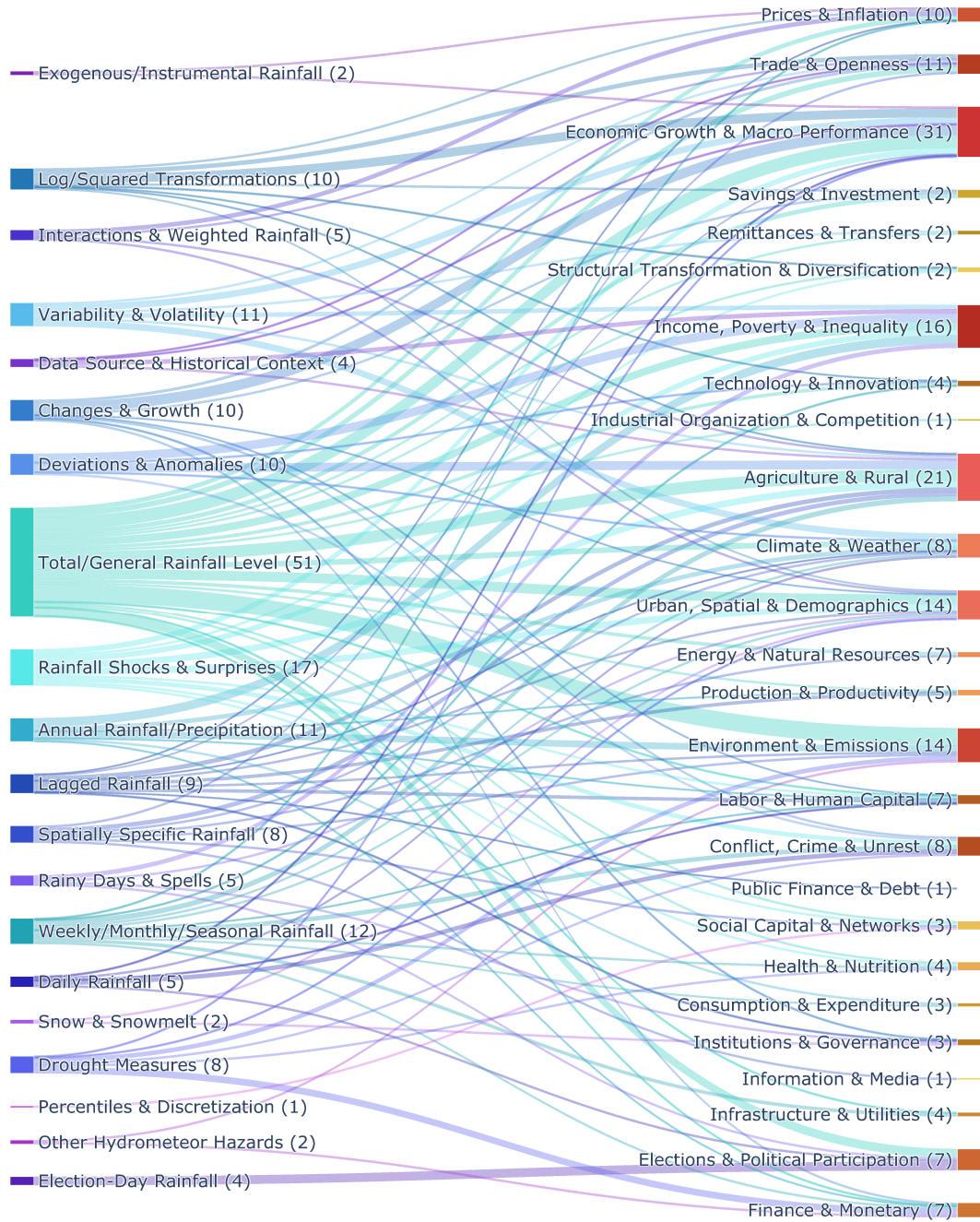
Note: The figure compares performance across zero-shot, retrieval-augmented generation (RAG), and supervised fine-tuning (SFT) using the common held-out evaluation set of $n = 88$ papers. Panel A reports confusion mosaics for the four binary extraction tasks, where cell area and labels indicate the number of true negatives (TN), false positives (FP), false negatives (FN), and true positives (TP). Panel B reports the corresponding accuracy, sensitivity, specificity, precision, F1-score, and balanced accuracy for each implementation and task. Panels C and D evaluate string extraction using semantic similarity between model-generated and target strings. Panel C reports the minimum-maximum range, mean $\pm$ one standard deviation, mean, and median for each field and implementation, with the corresponding sample size shown at right. Panel D plots the distribution of semantic similarity scores, with dashed vertical lines indicating implementation-specific means. For conditional string-extraction fields, semantic similarity is defined only for TP observations, so sample sizes vary across fields and implementations.

Figure 5: Rainfall Instruments and Endogenous Variables: Zero-Shot



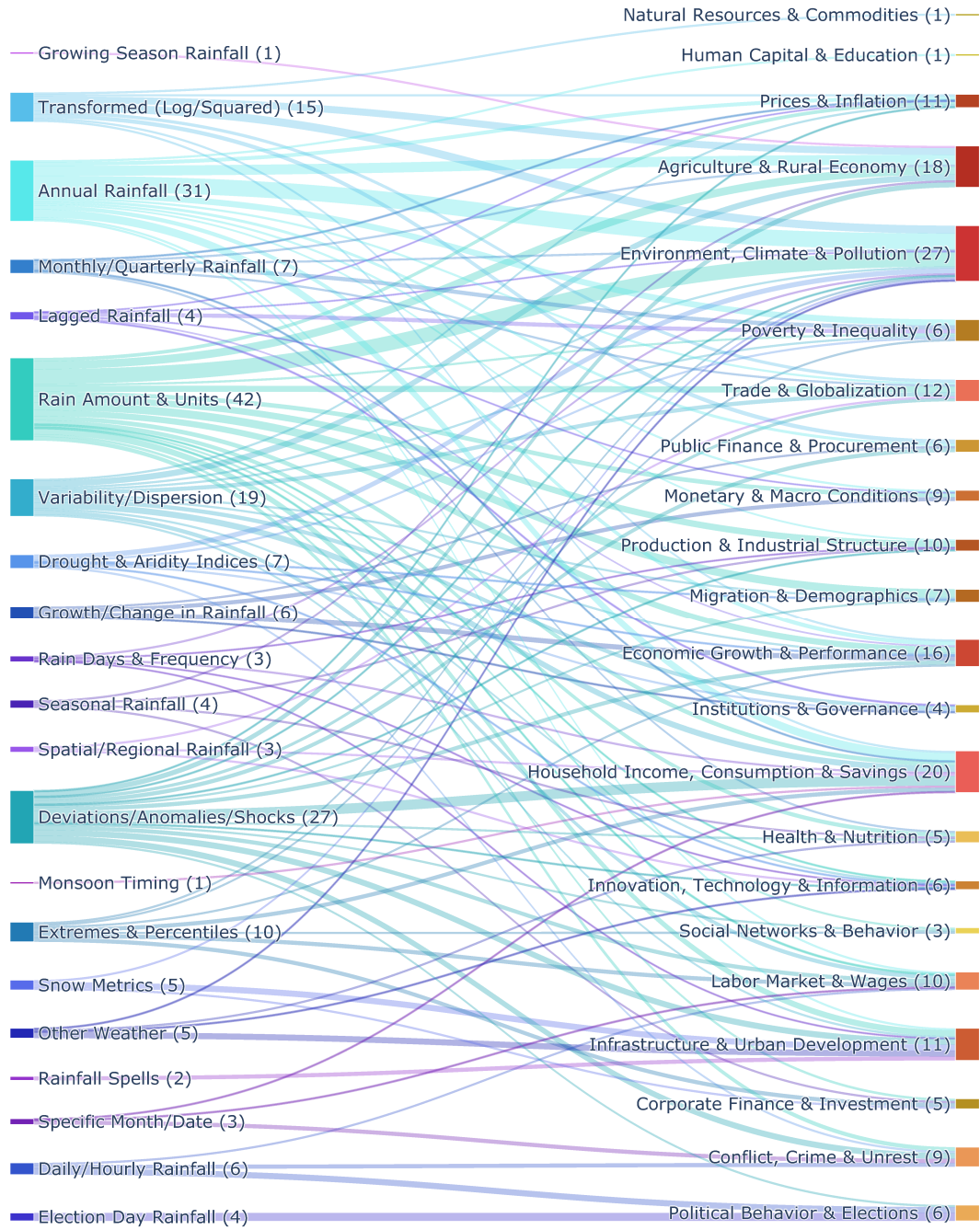
Note: The Sankey diagram maps harmonized categories of rainfall instruments identified by the zero-shot implementation (left) to the corresponding harmonized categories of endogenous variables (right). Flow width is proportional to the number of extracted rainfall instrument–endogenous variable linkages, with flows colored according to the rainfall-instrument category. Values in parentheses report the total number of extracted linkages associated with each category. Counts refer to model-generated linkages and need not correspond to unique papers.

Figure 6: Rainfall Instruments and Endogenous Variables: RAG



Note: The Sankey diagram maps harmonized categories of rainfall instruments identified by the retrieval-augmented generation (RAG) implementation (left) to the corresponding harmonized categories of endogenous variables (right). Flow width is proportional to the number of extracted rainfall instrument–endogenous variable linkages, with flows colored according to the rainfall-instrument category. Values in parentheses report the total number of extracted linkages associated with each category. Counts refer to model-generated linkages and need not correspond to unique papers.

Figure 7: Rainfall Instruments and Endogenous Variables: SFT



Note: The Sankey diagram maps harmonized categories of rainfall instruments identified by the supervised fine-tuning (SFT) implementation (left) to the corresponding harmonized categories of endogenous variables (right). Flow width is proportional to the number of extracted rainfall instrument–endogenous variable linkages, with flows colored according to the rainfall-instrument category. Values in parentheses report the total number of extracted linkages associated with each category. Counts refer to model-generated linkages and need not correspond to unique papers.

Table 1: Information Extracted from Each Paper and Prompt Dependencies

Measure	Information captured	Output	Dependency
Article Title	Article title, including subtitle	Exact string	None
DOI	Version-of-record DOI	DOI or n/a	None
Empirical Analysis	Presence of empirical statistical estimation	1/0	None
Dependent Variable(s)	Name(s) of main regression outcome(s)	String list	Empirical Analysis = 1
Endogeneity Problem	Main analysis has an endogenous regressor	1/0	Empirical Analysis = 1
Endogenous Variable(s)	Name(s) of endogenous regressor(s)	String list	Endogeneity Problem = 1
Instrumental Variable Regression	Use of an IV-type estimator with excluded instrument	1/0	Endogeneity Problem = 1
Instrumental Variable(s)	Name(s) of excluded instrument(s)	String list	Instrumental Variable Regression = 1
Rainfall Instrument	An excluded instrument is precipitation-based	1/0	Instrumental Variable Regression = 1
Rainfall Variable(s)	Name of specific precipitation-based instrument(s)	String list	Rainfall Instrument = 1

Note: Article title, DOI, and empirical analysis are evaluated independently. Dependent variables and the presence of an endogeneity problem are evaluated only for papers containing empirical analysis. Subsequent fields follow the conditional structure shown in the dependency column. A field is not evaluated when its dependency condition is not satisfied and it and all subsequent fields are coded **n/a**. String lists preserve the variable names used in the paper and separate multiple values with semicolons. Complete prompt instructions and output rules appear in Appendix 3.3.

Table 2: Binary Classification and Validation Metrics

Target Value	LLM-Generated Value		
	$Z_{ij} = 1$	$Z_{ij} = 0$	
$T_{ij} = 0$	FP : α_i	TN : $1 - \alpha_i$	Specificity = $\Pr(Z_{ij} = 0 \mid T_{ij} = 0) = 1 - \alpha_i$
$T_{ij} = 1$	TP : $1 - \beta_i$	FN : β_i	Sensitivity = $\Pr(Z_{ij} = 1 \mid T_{ij} = 1) = 1 - \beta_i$
Accuracy = $\Pr(Z_{ij} = T_{ij}) = (1 - \beta_i)p_i + (1 - \alpha_i)(1 - p_i)$, $p_i = \Pr(T_{ij} = 1)$			

Note: i indexes the extraction task and j indexes papers. α_i is the false-positive rate, β_i is the false-negative rate, and $p_i = \Pr(T_{ij} = 1)$ is the prevalence of the target value. Cell entries report the corresponding classification type and conditional probability.

Table 3: Cross-Implementation Agreement in Model-Labeled Rainfall-IV Records

<i>Panel A: Rainfall-IV identification</i>				
	Zero-shot	RAG	SFT	
Rainfall-IV papers	160	168	187	
Share of common model-labeled corpus (%)	4.17	4.38	4.87	
Unique to implementation	9	14	35	
<i>Panel B: Cross-implementation overlap in rainfall-IV identification</i>				
	Papers	Share of union (%)		
Identified by at least one implementation	218	100.0		
Identified by all three	137	62.8		
Identified by exactly two	23	10.6		
Identified by exactly one	58	26.6		
Any disagreement across implementations	81	37.2		
<i>Panel C: Exact agreement conditional on rainfall-IV identification by all three</i>				
	Zero-shot–RAG	Zero-shot–SFT	RAG–SFT	All three
Dependent variable(s)	57 (41.6)	6 (4.4)	7 (5.1)	5 (3.6)
Endogenous variable(s)	67 (48.9)	44 (32.1)	48 (35.0)	37 (27.0)
Rainfall instrument(s)	67 (48.9)	30 (21.9)	28 (20.4)	24 (17.5)
Rainfall instrument and endogenous variable lists	36 (26.3)	14 (10.2)	10 (7.3)	8 (5.8)

Note: The sample contains 3,840 unique papers with model-generated labels in all three files, including papers from the common held-out evaluation set. The 351 non-held-out human-reference records are excluded by filename. Repeated filenames retain their first occurrence in the original CSV order. Panel A counts explicit rainfall-IV labels of 1; all other codes are not counted as positive. Panel B reports overlap among the 218 papers labeled positive in at least one file. Panel C reports exact agreement among the 137 papers labeled positive in all three, with percentages of that sample in parentheses. Strings are lowercased, whitespace is standardized, and semicolon-separated lists are sorted without removing repeated items. Missing-value strings remain literal and count as agreement when identical. The final row requires agreement on both instrument and endogenous-variable lists, not their within-paper pairings. Exact-string disagreement need not imply disagreement in meaning.

Online-Only Appendix to “Mining Meaning: Measurement Error in Literature Reviews Using Generative AI”

A Operationalizing the Measurement Error Model

This appendix describes how to move from task-level validation of an LLM measurement system to an assessment of how measurement-system choice affects the substantive results of a literature review or any research related activity tasked to an LLM. The key distinction is that validation metrics characterize the quality of the model-generated data, whereas $\hat{\theta}_i^{(m)}$ is the result of applying a substantive analysis, g_i , to the data generated by model m . The measurement error metrics are the primitives while the substantive estimands capture the effects of measurement error on inference. There is thus no unique value of θ until the researcher specifies the descriptive quantity or statistical relationship of interest.

A.1 Task-Level Validation Metrics

Let $\mathcal{E}$ denote the evaluation sample, with $j \in \mathcal{E}$ a paper (or other object) in the evaluation sample. Additionally, let $A_i^{(m)}$, $R_i^{(m)}$, $C_i^{(m)}$, $\Pi_i^{(m)}$, $F1_i^{(m)}$, and $BA_i^{(m)}$ denote the population accuracy, sensitivity, specificity, precision, F1 score, and balanced accuracy and denote their evaluation-sample analogues as $\hat{A}_i^{(m)}$, $\hat{R}_i^{(m)}$, $\hat{C}_i^{(m)}$, $\hat{\Pi}_i^{(m)}$, $\hat{F1}_i^{(m)}$, and $\hat{BA}_i^{(m)}$. Then, for concept i and measurement system m , define

$$TP_i^{(m)} = \sum_{j \in \mathcal{E}} \mathbf{1} \left\{ T_{ij} = 1, Z_{ij}^{(m)} = 1 \right\}, \quad (\text{A.1})$$

$$FN_i^{(m)} = \sum_{j \in \mathcal{E}} \mathbf{1} \left\{ T_{ij} = 1, Z_{ij}^{(m)} = 0 \right\}, \quad (\text{A.2})$$

$$FP_i^{(m)} = \sum_{j \in \mathcal{E}} \mathbf{1} \left\{ T_{ij} = 0, Z_{ij}^{(m)} = 1 \right\}, \quad (\text{A.3})$$

$$TN_i^{(m)} = \sum_{j \in \mathcal{E}} \mathbf{1} \left\{ T_{ij} = 0, Z_{ij}^{(m)} = 0 \right\}. \quad (\text{A.4})$$

The corresponding binary validation measures are as follows.

Accuracy. Accuracy is the proportion of observations classified correctly:

$$\hat{A}_i^{(m)} = \frac{TP_i^{(m)} + TN_i^{(m)}}{TP_i^{(m)} + TN_i^{(m)} + FP_i^{(m)} + FN_i^{(m)}}. \quad (\text{A.5})$$

Accuracy summarizes overall classification performance, but it is prevalence dependent and does not distinguish FPs from FNs.

Sensitivity. Sensitivity, also called the TP rate or recall, is

$$\widehat{R}_i^{(m)} = \frac{\text{TP}_i^{(m)}}{\text{TP}_i^{(m)} + \text{FN}_i^{(m)}}. \quad (\text{A.6})$$

Sensitivity measures the proportion of target-positive observations correctly identified by the measurement system.

Specificity. Specificity, or the TN rate, is

$$\widehat{C}_i^{(m)} = \frac{\text{TN}_i^{(m)}}{\text{TN}_i^{(m)} + \text{FP}_i^{(m)}}. \quad (\text{A.7})$$

Specificity measures the proportion of target-negative observations correctly identified.

Precision. Precision, or the positive predictive value, is

$$\widehat{\Pi}_i^{(m)} = \frac{\text{TP}_i^{(m)}}{\text{TP}_i^{(m)} + \text{FP}_i^{(m)}}. \quad (\text{A.8})$$

Precision measures the proportion of model-generated positive classifications that are target positive.

F1 score. The F1 score is the harmonic mean of precision and sensitivity:

$$\widehat{\text{F1}}_i^{(m)} = \frac{2\widehat{\Pi}_i^{(m)}\widehat{R}_i^{(m)}}{\widehat{\Pi}_i^{(m)} + \widehat{R}_i^{(m)}}. \quad (\text{A.9})$$

The F1 score balances positive predictive performance and positive case recovery but does not incorporate true negatives.

Balanced accuracy. Balanced accuracy gives equal weight to positive and negative case performance:

$$\widehat{\text{BA}}_i^{(m)} = \frac{1}{2} \left(\widehat{R}_i^{(m)} + \widehat{C}_i^{(m)} \right). \quad (\text{A.10})$$

Balanced accuracy is useful when the positive and negative classes are highly unequal in size.

Semantic similarity. For a binary TP, let N_{ij} denote the target string and $X_{ij}^{(m)}$ the string generated by measurement system m . Semantic similarity is

$$S_{ij}^{(m)} = s\left(X_{ij}^{(m)}, N_{ij}\right) \in [0, 1] \quad \text{if } T_{ij} = Z_{ij}^{(m)} = 1. \quad (\text{A.11})$$

The validation-sample mean semantic similarity among true positives is

$$\hat{\mu}_{i,\mathcal{E}}^{(m)} = \frac{1}{\text{TP}_i^{(m)}} \sum_{j \in \mathcal{E}} \mathbf{1}\{T_{ij} = Z_{ij}^{(m)} = 1\} S_{ij}^{(m)}. \quad (\text{A.12})$$

Semantic similarity evaluates the quality of the generated string conditional on a correct positive classification. It does not penalize FNs, because no generated string exists in those cases, and it does not evaluate FPs, because no target string exists.

Omnibus Loss Function. The sample analogue of the task-level omnibus loss defined in the main text is

$$\begin{aligned} \hat{\mathcal{L}}_{i,\mathcal{E}}^{(m)} &= \frac{\text{FP}_i^{(m)} + \text{FN}_i^{(m)} + \sum_{j \in \mathcal{E}} \mathbf{1}\{T_{ij} = Z_{ij}^{(m)} = 1\} (1 - S_{ij}^{(m)})}{J_{\mathcal{E}}} \\ &= \left(1 - \hat{A}_i^{(m)}\right) + \frac{\text{TP}_i^{(m)}}{J_{\mathcal{E}}} \left(1 - \hat{\mu}_{i,\mathcal{E}}^{(m)}\right). \end{aligned} \quad (\text{A.13})$$

These statistics characterize task-level measurement performance. They do not, by themselves, determine the consequences of measurement error for the substantive quantities to be estimated. These statistics also cannot be interpreted as an error rate, because while the first term is a percentage value the third term in a semantic similarity score.

A.2 From Task-Level Binary Validation to Substantive Estimand

Let $r \in \{\mathcal{E}, P\}$ index either the evaluation sample or the full analysis sample or population, and let J_r denote the number of papers in r . Additionally, let $\mathbf{T}_{i,r}$ collect the target values for task i and let $\mathbf{Z}_{i,r}^{(m)}$ collect the corresponding values generated by measurement system m . Then define the target estimand and its model-generated analogue as

$$\theta_{i,r} = g_i(\mathbf{T}_{i,r}), \quad \hat{\theta}_{i,r}^{(m)} = g_i(\mathbf{Z}_{i,r}^{(m)}),$$

where g_i is the substantive analysis applied to concept i . Depending on the research question, $g_i(\cdot)$ may return a prevalence, mean, difference between groups, time trend, correlation, regression coefficient or even a string value like a variable name. The system-specific measurement-induced distortion is

$$\Delta_{i,r}^{(m)} = \widehat{\theta}_{i,r}^{(m)} - \theta_{i,r} \quad (\text{A.14})$$

This quantity is defined for either $r = \mathcal{E}$ or $r = P$ but it can only be directly calculated when the target values are observed, which will ordinarily be the case for only the evaluation sample.

The mean estimate across the M measurement systems is

$$\bar{\theta}_{i,r} = \frac{1}{M} \sum_{m=1}^M \widehat{\theta}_{i,r}^{(m)}. \quad (\text{A.15})$$

And measurement system dispersion is

$$D_{i,r}(\theta) = \left[\frac{1}{M} \sum_{m=1}^M \left(\widehat{\theta}_{i,r}^{(m)} - \bar{\theta}_{i,r} \right)^2 \right]^{1/2}. \quad (\text{A.16})$$

Because the measurement systems constitute a fixed, prespecified collection of models and implementations rather than a random sample from a population of systems, the denominator is M . Similarly, $\bar{\theta}_{i,r}$ is the center of the estimates produced by the selected measurement systems. It is not necessarily closer to the target estimand than any particular system-specific estimate and should not be interpreted as an unbiased measurement system-average estimator without additional assumptions. Nor is the quantity $D_{i,r}(\theta)$ a standard error or confidence interval. It is a descriptive measure of the sensitivity of the substantive result to choice of measurement system.

Example 1: Prevalence. Suppose the substantive estimand for concept i is the proportion of papers in which that concept is present. In this case, the function g_i is the sample mean, so that

$$\theta_r = \frac{1}{J} \sum_{j \in r} T_j, \quad \widehat{\theta}_r^{(m)} = \frac{1}{J} \sum_{j \in r} Z_j^{(m)}.$$

In the notation of the binary measurement model, θ_r is the target prevalence p , $\widehat{\theta}_r^{(m)}$ is the measured prevalence $q^{(m)}$, and $\Delta_r^{(m)} = q^{(m)} - p$.

For example, if the target prevalence in the evaluation sample was $\theta_{\mathcal{E}} = 0.10$ and four measurement systems produced prevalence estimates

$$0.10, \quad 0.12, \quad 0.11, \quad 0.15.$$

Then, the measurement systems produce a mean estimate of 12%, with absolute bias of 2 percentage points and relative bias of 20%, with cross-system dispersion of approximately 1.9 percentage points.

Example 2: Numerical Measures and Regression Relationships. The same procedure applies when binary outputs are combined into a numerical summary. Examples include the number

of hypotheses tested in a paper, the number of variables assigned a particular econometric role, or the share of a prespecified set of concepts, like JEL codes, detected in each paper. Let $Y_j^{(m)}$ denote the numerical summary constructed from the data generated by measurement system m .

Suppose, for example, that the researcher is interested in the relationship between this numerical summary and a paper characteristic, W_j , such as year of publication. For each measurement system, estimate the same regression:

$$Y_j^{(m)} = a^{(m)} + \theta^{(m)} W_j + \mathbf{B}_j' \boldsymbol{\gamma}^{(m)} + \varepsilon_j^{(m)}, \quad (\text{A.17})$$

where $a^{(m)}$ is the intercept and $\mathbf{B}_j$ contains any additional paper characteristics included in the analysis. The estimated coefficient on W_j is the system-specific substantive estimate $\hat{\theta}^{(m)}$. The researcher then can calculate the cross-system mean and measurement-system dispersion.

The same logic applies if the LLM-generated variable enters on the right-hand side of a regression, or if $g_i(\cdot)$ is a nonlinear estimator. The relevant requirement is that the same specification, sample restrictions, covariates, transformations, and estimation procedure be applied to every model-generated dataset. Each $\hat{\theta}_{i,r}^{(m)}$ may also have a conventional sampling standard error when such an interpretation is supported by the sampling design. That standard error measures within-system uncertainty, whereas $D_{i,r}(\theta)$ measures between-system sensitivity.

A.3 From Task-Level String Validation to Substantive Estimand

Let $\mathbf{N}_{i,r}$ collect the target strings for task i and let $\mathbf{X}_{i,r}^{(m)}$ collect the corresponding strings generated by measurement system m . Because strings do not have a natural arithmetic structure, a string based analysis must first map the strings into a scalar descriptive quantity. Then define the target estimand and the model-generated estimand as

$$\theta_{i,r} = g_i(\mathbf{T}_{i,r}, \mathbf{N}_{i,r}), \quad \hat{\theta}_{i,r}^{(m)} = g_i(\mathbf{Z}_{i,r}^{(m)}, \mathbf{X}_{i,r}^{(m)}),$$

The target indicators and strings appear together because a target string is only defined when $T_{i,r} = 1$ and a generated string is only defined when $Z_{i,r}^{(m)} = 1$.

The system-specific measurement-induced distortion, the mean estimate across the measurement systems, and the measurement system dispersion are all defined analogous to the binary outcome case. The system-specific estimate, cross-system mean, and measurement-system dispersion can be calculated for the full sample once the generated strings have been transformed into a common scalar representation.

Example 1: Categorization. Raw strings can be mapped onto a common analytic representation using a fixed harmonization rule. Let

$$h(\cdot) : \mathcal{X} \rightarrow \{1, \dots, K\} \quad (\text{A.18})$$

map each target or generated strings into one of K standardized concepts or categories.

For example, suppose task i extracts the name of an instrumental variable and K is a set of standardized economic concepts, such as JEL codes. Define the target and generated category indicators as

$$\theta_{k,r} = \frac{1}{J_r} \sum_{j \in r} \mathbf{1}\{T_j = 1, h(N_j) = k\}, \quad \widehat{\theta}_{k,r}^{(m)} = \frac{1}{J_r} \sum_{j \in r} \mathbf{1}\{Z_j^{(m)} = 1, h(X_j^{(m)}) = k\}.$$

Thus, $\theta_{k,r}$ is the share of papers whose target string belongs to category k , while $\widehat{\theta}_{k,r}^{(m)}$ is the corresponding share measured using the strings generated by measurement system m .

The denominator should be chosen to match the estimand. Dividing by all papers estimates the share of papers using a variable in category k . Dividing by the number of model-positive observations instead estimates the share of extracted variables assigned to category k . These are distinct descriptive quantities.

The harmonization rule $h(\cdot)$ must be held fixed across measurement systems. If harmonization is itself performed by an LLM or another learned model, that procedure forms part of the measurement system and should be included in the definition of m .

Example 2: Semantic Proximity to a Category. A researcher may be interested not only in whether an extracted string belongs to a discrete category but also how closely a string maps to a substantive concept. Let $b_k \in \mathcal{X}$ denote a prespecified reference string representing concept k . For example, if task i extracts the name of an instrumental variable, b_k might be the reference string “weather shock.”

Using the same normalized semantic similarity function, $s(\cdot, \cdot)$, define the target mean semantic proximity and its model-generated analog as

$$\theta_{k,r} = \frac{1}{J_r} \sum_{j \in r, T_j=1} s(N_j, b_k), \quad \widehat{\theta}_{k,r}^{(m)} = \frac{1}{J_r} \sum_{j \in r, Z_j^{(m)}=1} s(X_j^{(m)}, b_k).$$

A paper contributes zero when no target or generated string is present and contributes a value between zero and one when a string is present. Thus, $\theta_{k,r}$ is the average semantic proximity of the target strings to concept k , while $\widehat{\theta}_{k,r}^{(m)}$ is the corresponding quantity generated by measurement system m .

Alternatively, the estimands can be written as

$$\theta_{k,r} = p_r \bar{s}_{k,r} \quad \hat{\theta}_r^{(m)} = \hat{q}_r^{(m)} \bar{s}_{k,r}^{(m)},$$

where p_r is the share of papers for which the target string exists, $\hat{q}_r^{(m)}$ is the share of papers for which measurement system m produces a string, and $\bar{s}_{k,r}$ is the mean semantic proximity to b_k conditional on a string being produced. This estimand combines an extensive margin (whether a string exists) with an intensive margin (how closely that string relates to the reference concept).

However, care must be taken in interpreting this extensive/intensive margin estimand. For example, if the target strings exist in 40% of papers in the evaluation sample with mean semantic proximity to $b_k = 0.60$, then the resulting estimand is $\theta_{k,\mathcal{E}} = 0.24$. Suppose measurement system m produces a string in 30% of papers in the evaluation sample and with mean semantic proximity to $b_k = 0.80$. Then $\hat{\theta}_{k,\mathcal{E}} = 0.24$, the same overall estimate despite different rates of prevalence and semantic proximity. Therefore, researchers should report the prevalence and conditional semantic proximity components alongside the substantive estimands.

As in all the examples, the system-specific estimates $\left(\hat{\theta}_{i,r}^{(m)}\right)$, cross-system means $(\bar{\theta}_{i,r})$, and measurement-system dispersion $(D_{i,r}(\theta))$ can be calculated for both the evaluation sample ($\mathcal{E}$) and the full sample or population (P). The absolute bias $\left(q_{i,\mathcal{E}}^{(m)} - p_{i,\mathcal{E}}\right)$, relative bias $\left(\left[q_{i,\mathcal{E}}^{(m)} - p_{i,\mathcal{E}}\right] / p_{i,\mathcal{E}}\right)$, target estimand $(\theta_{i,\mathcal{E}})$, and system-specific distortion $\left(\Delta_{i,\mathcal{E}}^{(m)}\right)$ can only be calculate where the corresponding target values are observed, which is typically only in the evaluation sample.

A.4 Evaluation Sample and Full Sample

The evaluation sample and full sample support different empirical quantities.

Evaluation Sample. Because both target and generated values are observed, the evaluation sample supports calculation of

- confusion-matrix quantities, including accuracy, sensitivity, specificity, precision, F1, and balanced accuracy;
- semantic similarity and the omnibus measurement loss;
- the target-data estimand, $\theta_{i,\mathcal{E}}$;
- each system-specific estimand, $\hat{\theta}_{i,\mathcal{E}}^{(m)}$;
- direct estimand distortion, $\Delta_{i,\mathcal{E}}^{(m)}$;
- the cross-system mean, $\bar{\theta}_{i,\mathcal{E}}$, and measurement-system dispersion, $D_{i,\mathcal{E}}(\theta)$.

The evaluation sample therefore reveals both how accurately each task is performed and how the resulting errors affect the particular descriptive estimand.

Full sample. In the full sample, the target values generally remain unobserved, meaning the performance metrics cannot be directly calculated. Researchers can nevertheless calculate $\widehat{\theta}_{i,P}^{(m)}$ for every measurement system along with $\bar{\theta}_{i,P}$ and $D_{i,P}(\theta)$. Without target labels, the full-sample estimand distortion $\Delta_{i,P}^{(m)}$ cannot be calculated.

The interpretation of full-sample dispersion is therefore narrower: it measures dependence on measurement-system choice, not absolute error. Low dispersion does not establish validity because all systems may make similar errors. High dispersion indicates that substantive conclusions are sensitive to the system used to generate the data but does not, by itself, identify which estimate is closest to the target.

Any attempt to use validation-sample error rates to correct full-sample estimates requires that measurement performance in the evaluation sample transport to the full sample. This assumption is more credible when validation observations are representative of the full corpus or when validation is stratified over paper characteristics associated with measurement difficulty.

A.5 Practical Workflow

A researcher implementing this framework can proceed as follows.

1. **Define the target and estimand.** Specify each target concept T_{ij} and the substantive quantity $\theta_i = g_i(\mathbf{T}_i)$ before evaluating measurement system performance. Examples include prevalence, a difference between periods, a time trend, or a regression coefficient.
2. **Fix the measurement systems.** Define each measurement system m to include the model, model version, prompts, retrieval procedure, decoding settings, parser, and any string-harmonization procedure.
3. **Use a common set of papers.** Apply every measurement system to the same validation and analysis observations. Otherwise, cross-system differences may reflect differences in sample composition rather than measurement-system choice.
4. **Construct an expert-labeled evaluation sample.** Record the target binary values and, when applicable, target strings using a fixed coding and adjudication protocol.
5. **Calculate task-level validation measures.** For each concept and measurement system, calculate the confusion matrix, accuracy, sensitivity, specificity, precision, F1, balanced accuracy, mean semantic similarity, and omnibus measurement loss.
6. **Apply the substantive analysis to the validation data.** Calculate $\theta_{i,\mathcal{E}}$ and $\widehat{\theta}_{i,\mathcal{E}}^{(m)}$ for every measurement system. Report the system-specific distortions, $\Delta_{i,\mathcal{E}}^{(m)}$.
7. **Apply the same analysis to the full sample.** Holding the specification fixed, calculate $\widehat{\theta}_{i,P}^{(m)}$ for every measurement system.

8. **Summarize measurement-system sensitivity.** Calculate and report each system-specific estimate, the cross-system mean, $\bar{\theta}_{i,P}$, and measurement-system dispersion, $D_{i,P}(\theta)$. Do not interpret the cross-system mean as the target value or the dispersion as a confidence interval.
9. **Keep sampling uncertainty and measurement-system sensitivity separate.** Report conventional standard errors or confidence intervals for each system-specific estimate when justified by the sampling design. Report $D_i(\theta)$ separately as a measure of sensitivity to the measurement system.
10. **Preserve dependence when quantifying uncertainty.** If uncertainty intervals are constructed for validation metrics, estimand distortion, or measurement-system dispersion, re-sample papers jointly across all measurement systems. Models should not be resampled, because they constitute the fixed set being compared.

The central operational principle is simple: define the substantive analysis once, apply it unchanged to every model-generated dataset, and compare the resulting estimates. Task-level validation shows how accurately the data are measured; estimand distortion and measurement-system dispersion show whether those errors matter for the conclusions drawn from model outputs.

B Additional Details on Implementation Pipeline

Figure B.1 provides a visual description of the research design pipeline. Panel 1 shows the collection of 4,191 candidate papers and human labeling of a subset of 439 papers. Of the labeled sample, 20% forms the common held-out evaluation set and 80% supplies reference data for retrieval and fine-tuning. Panel 2 distinguishes implementation inputs. Zero-shot receives no labeled examples, RAG retrieves reference examples, and SFT uses labeled data for training and development monitoring. All implementations receive the focal paper and fixed prompt context. Panel 3 compares outputs with adjudicated labels on the common held-out set and produces a candidate corpus output dataset for each measurement system. Panel 4 applies the same substantive analysis to each generated dataset and compares the resulting estimands.

B.1 Fixed Context

To promote consistency for our repetitive metadata-extraction task across a sequence of queries and model implementations, we make use of a fixed prompting structure which appears at the beginning of each API call to condition model behavior. A model is first provided a system role, which establishes its task role and defines output constraints. This is followed by a user role, which defines the specific extraction task and provides the model with the exact context from which it is expected to identify the requested metadata. Finally, an assistant role (prefill) acknowledges these defined constraints and initializes the interaction. This prompting strategy does not eliminate per-paper model discretion, but helps constrain its judgment to supplied evidence and an expected response schema which can be repurposed across all implementations to improve comparability.

System Role

```
{  "role": "system", "content": (  
    "You are an AI assistant that is an expert in analysis of economic literature."  
    "You interpret complex content and extract specific information, especially about"  
    "empirical methods, endogeneity problems, and instrumental variable strategies."  
    "Rules: Use only information from the provided text to answer each query."  
    "If the requested information is not available, answer exactly n/a."  
    "Follow the requested output format exactly."  
    "Exclusions: Do not include the user queries, any labels, greek letters, or additional"  
    "text beyond what is requested. Do not respond with symbolic notation, only words."  
    "For binary questions with justification requests, always start with 0 or 1" },
```

User Role

```
{  "role": "User", "content": (  
    "You will be asked a sequence of extraction questions about the same academic article. "  
    "I am specifically interested in how researchers address endogeneity problems, particularly "  
    "using instrumental variables and especially rainfall-based instruments. "  
    "Use answers you have already given as context for later questions when helpful, "  
    "but always ground your answers in the provided text."  
    "Here are the relevant sections from the article:"  
    "Rainfall Shocks and Smallholder Market Participation in Rural Malawi ..."  
    "We estimate the effect of market participation on household income using a two-stage"  
    "least squares strategy. Because market participation is plausibly endogenous to income..."  
  ),
```

Assistant Role (Prefill)

```
{  "role": "Assistant", "content": (  
    "Understood. I will answer each extraction question using only the provided article text."  
  ),
```

B.2 Model Queries

Article Title

```
{  'key': 'Article Title',  
  'question': 'Task: Extract the article title. Look: top of first page (main heading  
before authors/abstract). Rules: include subtitle if in the same heading; exclude au-  
thors/affiliations/journal headers/footers/section headers; ignore footnote markers (*, †, su-  
perscripts). Output: ONE line: title text only. No quotes, labels, or extra words. — Task:  
Extract the article title. Look: top of first page (main heading before authors/abstract).  
Rules: include subtitle if in the same heading; exclude authors/affiliations/journal head-  
ers/footers/section headers; ignore footnote markers (*, †, superscripts). Output: ONE line:  
title text only. No quotes, labels, or extra words.' },
```

DOI

```
{ 'key': 'DOI',  
  'question': 'Task: Extract the DOI of THIS article (version of record), not DOIs in  
references. Look: first page/front matter for doi:, DOI, https://doi.org/. Rules: remove  
URL/prefix (doi:, https://doi.org/); remove spaces/line breaks; strip trailing punctuation;  
output lowercase. Output: ONE token = normalized DOI (10.xxxx/xxxx) OR exactly n/a.  
— Task: Extract the DOI of THIS article (version of record), not DOIs in references. Look:  
first page/front matter for doi:, DOI, https://doi.org/. Rules: remove URL/prefix (doi:,  
https://doi.org/); remove spaces/line breaks; strip trailing punctuation; output lowercase.  
Output: ONE token = normalized DOI (10.xxxx/xxxx) OR exactly n/a.' }
```

Empirical Analysis

```
{ 'key': 'Empirical Analysis',  
  'question': 'Task: Does the article contain empirical quantitative statistical estimation  
(e.g., regressions/econometrics with estimated coefficients/SEs/p-values)? Look: 'we esti-  
mate/regress', model equations with error terms, regression tables with coefficients. Rules:  
do NOT infer from topic/title/abstract. Count only analysis done in THIS article (not  
summaries or reviews of other papers). Exclude review articles, articles in hydrology, and  
articles that contain purely theoretical work only, qualitative/descriptive only, or numerical  
simulation or model calibration only. Output: 1 if yes, 0 if no. Output exactly one character.  
— Task: Does the article contain empirical quantitative statistical estimation (e.g., regres-  
sions/econometrics with estimated coefficients/SEs/p-values)? Look: 'we estimate/regress',  
model equations with error terms, regression tables with coefficients. Rules: do NOT in-  
fer from topic/title/abstract. Count only analysis done in THIS article (not summaries or  
reviews of other papers). Exclude review articles, articles in hydrology, and articles that  
contain purely theoretical work only, qualitative/descriptive only, or numerical simulation  
or model calibration only. Output: 1 if yes, 0 if no. Output exactly one character.' },
```

Dependent Variable(s)

```
{ 'key': 'Dependent Variable(s)',  
  'dependency': 'key': 'Empirical Analysis', 'value': '1',  
  'question': 'Task: List the main dependent/outcome variable(s) used in the primary regression/econometric results. Look: LHS of main equations; column headers of main regression tables; text describing the main empirical model. Rules: exclude first-stage outcomes, RHS variables (treatments/endogenous regressors/instruments/controls/covariates/fixed effects), mediators/moderators. Keep names exactly as written in the paper/table (including any log/ln/differences/units if shown). Output: ONE line: variable name(s) only; separate multiple with '; '. Output the name of the dependent variable, not the symbolic notation representing the name of the variable in an equation (e.g.,  $y_{it}$ ). — Task: List the main dependent/outcome variable(s) used in the primary regression/econometric results. Look: LHS of main equations; column headers of main regression tables; text describing the main empirical model. Rules: exclude first-stage outcomes, RHS variables (treatments/endogenous regressors/instruments/controls/covariates/fixed effects), mediators/moderators. Keep names exactly as written in the paper/table (including any log/ln/differences/units if shown). Output: ONE line: variable name(s) only; separate multiple with '; '. Output the name of the dependent variable, not the symbolic notation representing the name of the variable in an equation (e.g.,  $y_{it}$ ).' },
```

Endogeneity Problem

```
{ 'key': 'Endogeneity Problem',  
  'dependency': 'key': 'Empirical Analysis', 'value': '1',  
  'question': 'Task: In the MAIN empirical analysis, do the authors treat any RHS variable  
as endogenous (correlated with the error term) and address it explicitly? Look: statements  
that a regressor is endogenous + a method to address it (IV/2SLS/3SLS/LIML, control  
function/2SRI, GMM with instruments, first stage, weak-IV tests, etc.). Rules: do NOT  
count generic mentions of 'endogeneity' or references to endogeneity issues in other papers  
without an actual endogenous regressor in the main specs of this paper. Output: 1 if yes,  
0 if no. Output exactly one character. — Task: In the MAIN empirical analysis, do the  
authors treat any RHS variable as endogenous (correlated with the error term) and address  
it explicitly? Look: statements that a regressor is endogenous + a method to address it  
(IV/2SLS/3SLS/LIML, control function/2SRI, GMM with instruments, first stage, weak-  
IV tests, etc.). Rules: do NOT count generic mentions of 'endogeneity' or references to  
endogeneity issues in other papers without an actual endogenous regressor in the main specs  
of this paper. Output: 1 if yes, 0 if no. Output exactly one character.' },
```

Endogenous Variable(s)

```
{ 'key': 'Endogenous Variable(s)',  
  'dependency': 'key': 'Endogeneity Problem', 'value': '1',  
  'question': 'Task: List the explanatory variable(s) explicitly treated as endogenous in the  
main analysis. Look: 'we instrument X', 'X is endogenous', first-stage descriptions/tables,  
reduced form, weak-IV/overid tests. Rules: exclude instruments, dependent variables,  
and ordinary controls. Keep names exactly as written in the paper/table (including any  
log/ln/differences/units if shown). Output: ONE line: variable name(s) only; separate  
multiple with '; '. Output the name of the endogenous variable, not the symbolic notation  
representing the name of the variable in an equation (e.g.,  $x_{it}$ ). — Task: List the explanatory  
variable(s) explicitly treated as endogenous in the main analysis. Look: 'we instrument X', 'X  
is endogenous', first-stage descriptions/tables, reduced form, weak-IV/overid tests. Rules:  
exclude instruments, dependent variables, and ordinary controls. Keep names exactly as  
written in the paper/table (including any log/ln/differences/units if shown). Output: ONE  
line: variable name(s) only; separate multiple with '; '. Output the name of the endogenous  
variable, not the symbolic notation representing the name of the variable in an equation  
(e.g.,  $x_{it}$ ).' },
```

Instrumental Variable Regression

```
{ 'key': 'Instrumental Variable Regression',  
  'dependency': 'key': 'Endogeneity Problem', 'value': '1',  
  'question': 'Task: In the MAIN empirical analysis, do the authors implement an IV-type  
estimator with excluded instruments to address endogeneity? Look: an actual excluded in-  
strument set used in a first stage/reduced form; 2SLS/IV/3SLS/LIML; IV-Probit; control  
function/2SRI; GMM with instruments; exclusion restriction; first-stage equations/tables.  
Rules: do NOT count non-statistical uses of 'instrument' (survey instrument, measurement  
instrument), references to other papers that us IVs, arguments as to why IVs are not neces-  
sary for this paper, or papers that only use RCT/RDD/DiD/event study without an IV first  
stage. Output: 1 if yes, 0 if no. Output exactly one character. — Task: In the MAIN empir-  
ical analysis, do the authors implement an IV-type estimator with excluded instruments to  
address endogeneity? Look: an actual excluded instrument set used in a first stage/reduced  
form; 2SLS/IV/3SLS/LIML; IV-Probit; control function/2SRI; GMM with instruments; ex-  
clusion restriction; first-stage equations/tables. Rules: do NOT count non-statistical uses of  
'instrument' (survey instrument, measurement instrument), references to other papers that  
us IVs, arguments as to why IVs are not necessary for this paper, or papers that only use  
RCT/RDD/DiD/event study without an IV first stage. Output: 1 if yes, 0 if no. Output  
exactly one character.' },
```

Instrumental Variable(s)

```
{ 'key': 'Instrumental Variable(s)',  
  'dependency': 'key': 'Instrumental Variable Regression', 'value': '1',  
  'question': 'Task: List the excluded instrument(s) used in the main IV analysis. Look: 'we  
instrument X with Z', 'Z is our instrument', 'excluded instrument', first-stage/reduced-form  
equations or tables. Rules: exclude endogenous regressors themselves, dependent variables,  
and regular controls. Keep names exactly as written in the paper/table (including any  
log/ln/differences/units if shown). Output: ONE line: instrument name(s) only; separate  
multiple with '; '. Output the name of the instrumental variable, not the symbolic notation  
representing the name of the variable in an equation (e.g.,  $z_{it}$ ). — Task: List the excluded  
instrument(s) used in the main IV analysis. Look: 'we instrument X with Z', 'Z is our instru-  
ment', 'excluded instrument', first-stage/reduced-form equations or tables. Rules: exclude  
endogenous regressors themselves, dependent variables, and regular controls. Keep names  
exactly as written in the paper/table (including any log/ln/differences/units if shown). Out-  
put: ONE line: instrument name(s) only; separate multiple with '; '. Output the name of  
the instrumental variable, not the symbolic notation representing the name of the variable  
in an equation (e.g.,  $z_{it}$ ).' },
```

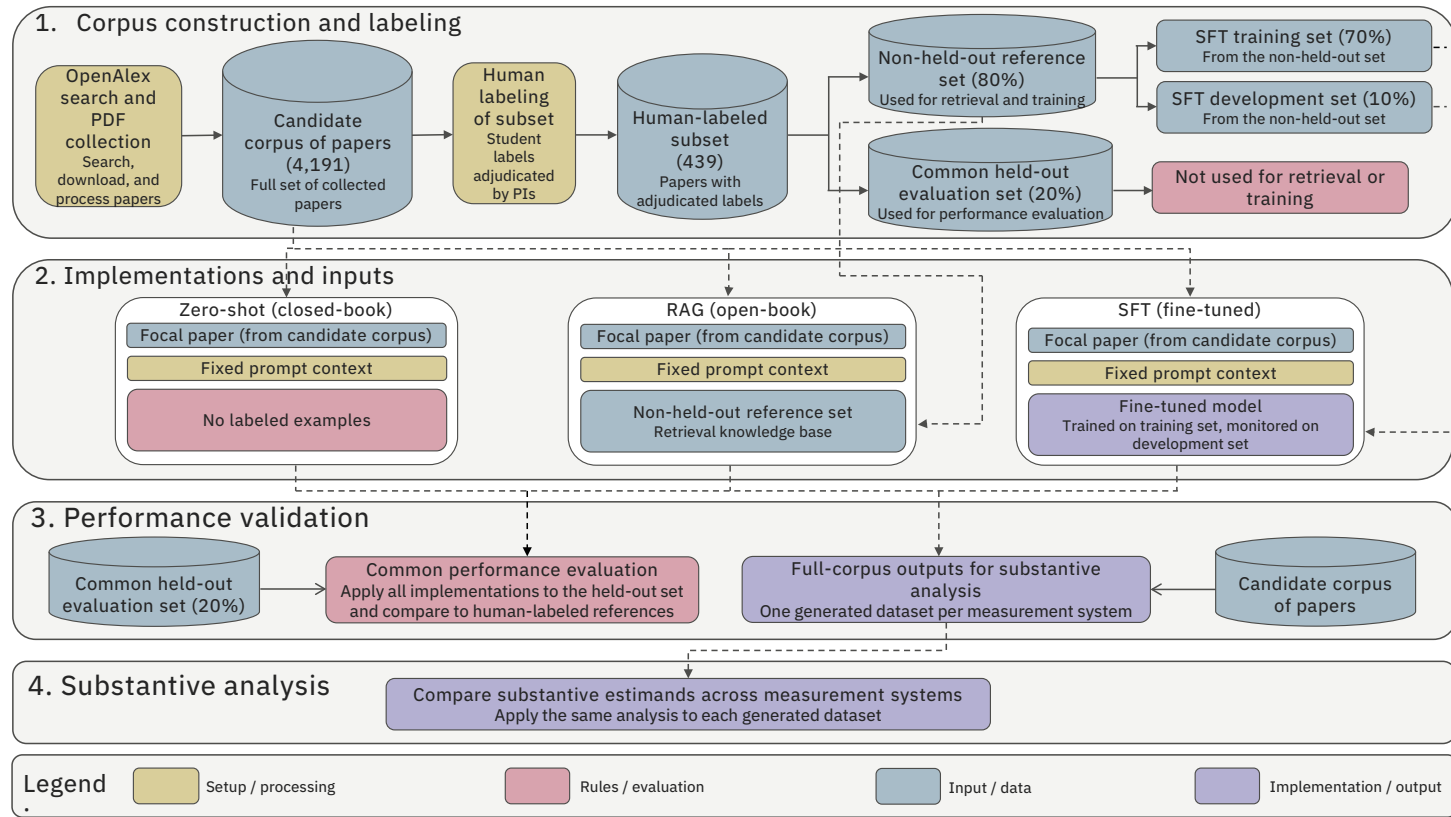
Rainfall Instrument

```
{ 'key': 'Rainfall Instrument',  
  'dependency': 'key': 'Instrumental Variable Regression', 'value': '1',  
  'question': 'Task: Is any excluded instrument based on rainfall/precipitation? Look:  
rainfall, precipitation, drought, monsoon rainfall/onset, wet-day counts, SPI/SPEI/PDSI,  
precipitation-derived indices used as the EXCLUDED instrument. Rules: count  
only if the variable is precipitation-based and used as an excluded instrument in  
an IV first stage/reduced form. Do NOT count precipitation used only as regres-  
sor/control/interaction/exposure/outcome or if its relationship to the dependent variable  
is discussed. Do NOT count ENSO or other climate indices unless explicitly stated to be  
precipitation-based AND used as the excluded instrument. Ignore mentions of rainfall as an  
instrument in other papers. Output: 1 if yes, 0 if no. Output exactly one character. —  
Task: Is any excluded instrument based on rainfall/precipitation? Look: rainfall, precip-  
itation, drought, monsoon rainfall/onset, wet-day counts, SPI/SPEI/PDSI, precipitation-  
derived indices used as the EXCLUDED instrument. Rules: count only if the variable is  
precipitation-based and used as an excluded instrument in an IV first stage/reduced form.  
Do NOT count precipitation used only as regressor/control/interaction/exposure/outcome  
or if its relationship to the dependent variable is discussed. Do NOT count ENSO or other  
climate indices unless explicitly stated to be precipitation-based AND used as the excluded  
instrument. Ignore mentions of rainfall as an instrument in other papers. Output: 1 if yes,  
0 if no. Output exactly one character.' },
```

Rainfall Variable(s)

```
{ 'key': 'Rainfall Variable(s)',  
  hspace*0.5cm'dependency': 'key': 'Rainfall Instrument', 'value': '1',  
  'question': 'Task: List the specific rainfall/precipitation-based excluded instrument(s)  
used in the main IV analysis. Look: first-stage/reduced-form equations/tables for the  
precipitation-based instrument name(s) (e.g., total/mean rainfall over a window, rainfall  
deviations/shocks, monsoon onset, rainfall index, coefficient of variation of rainfall). Rules:  
keep names exactly as written in the paper/table (including window/statistic/units/logs if  
shown). Output: ONE line: rainfall instrument name(s) only; separate multiple with ' ; '.  
Output the name of the rainfall instrument, not the symbolic notation representing the name  
of the variable in an equation (e.g.,  $r_{it}$ ). — Task: List the specific rainfall/precipitation-  
based excluded instrument(s) used in the main IV analysis. Look: first-stage/reduced-form  
equations/tables for the precipitation-based instrument name(s) (e.g., total/mean rainfall  
over a window, rainfall deviations/shocks, monsoon onset, rainfall index, coefficient of vari-  
ation of rainfall). Rules: keep names exactly as written in the paper/table (including win-  
dow/statistic/units/logs if shown). Output: ONE line: rainfall instrument name(s) only;  
separate multiple with ' ; '. Output the name of the rainfall instrument, not the symbolic  
notation representing the name of the variable in an equation (e.g.,  $r_{it}$ ).' }, ]
```

Figure B.1: Overall Research Design



Note: The figure summarizes corpus construction, implementation inputs, and evaluation. Panel 1 shows the collection of 4,191 candidate papers and human labeling of a subset of 439 papers. Of the labeled sample, 20% forms the common held-out evaluation set and 80% supplies reference data for retrieval and fine-tuning. Panel 2 distinguishes implementation inputs. Zero-shot receives no labeled examples, RAG retrieves reference examples, and SFT uses labeled data for training and development monitoring. All implementations receive the focal paper and fixed prompt context. Panel 3 compares outputs with adjudicated labels on the common held-out set and produces a full-corpus dataset for each measurement system. Panel 4 applies the same substantive analysis to each generated dataset and compares the resulting estimands.